SKILLFORGE

SOFTWARE REQUIREMENTS SPECIFICATION

Chapter 1 — Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define and document the requirements of the SkillForge prototype. The document provides a clear reference for the development, testing, evaluation, and demonstration of the proposed interactive puzzle-based skill development experience.

This SRS describes the expected behaviour, major features, user interactions, interface requirements, system requirements, and other relevant requirements of the prototype. It establishes a common understanding of the project requirements for the project team and provides a structured basis for implementing and evaluating the prototype.

1.2 Scope

Scope of the Project

SkillForge is an interactive puzzle-based skill development prototype designed around an ancient-futuristic environment and an environment-driven gameplay experience. The prototype focuses on a fixed-camera, 2.5D presentation in which the user interacts with the HUB environment, interface panels, puzzle mechanisms, and progression elements rather than freely exploring a conventional open-world environment.

The scope of the prototype includes the complete introductory experience, beginning with the opening sequence and progressing through the Hall Awakening and Guest HUB. The prototype includes the login/sign-up flow, data feed, glyph conversion, Forge Vision progression, challenge selection, challenge chamber, puzzle interaction, portal activation, reward presentation, and return to the HUB.

The challenge experience includes the Forge Vision requirement, Ring System, Alignment Analysis, Energy Flow, Core Resonance, Portal Ignition, and reward presentation through the Portal Warp. A separate Reward Chamber is not included in the prototype.

The prototype also covers progression and supporting systems such as save behaviour, notifications, hints, settings, interface interactions, audio feedback, and the required visual transitions. Progress is saved after challenge completion, while an incomplete challenge does not retain progress according to the defined save rules.

Out of Scope

The current prototype does not aim to provide a complete commercial-scale game or an unrestricted open-world experience. The SRS will document only the features and behaviour required for the approved prototype. Features outside the approved prototype scope will not be treated as current requirements.

The SRS will also avoid implementation-level coding details and will not introduce new design decisions during documentation. Every functional requirement documented for the prototype must be demonstrable in the final Unity prototype.

1.3 Overview

This Software Requirements Specification is organized to provide a structured description of the SkillForge prototype and its requirements. Chapter 1 introduces the purpose, scope, and overall organization of the project. **Chapter 2 provides a general description of the prototype, including its perspective, major**

functions, users, operating environment, constraints, and
assumptions.

Chapter 3 defines the functional requirements of the prototype, covering the opening Sequence, HUB, login and sign-up flow, data feed, glyph conversion, Forge progression, challenge system, puzzle interactions, portal activation, rewards, save behaviour, notifications, settings, and return to the HUB. Chapter 4 describes the user interface requirements, including the HUB and challenge interfaces, panels, notifications, hints, settings, screen flow, panel behaviour, and software interfaces.

The remaining chapters describe performance requirements, non-functional requirements, system architecture, data management, testing requirements, and acceptance criteria. The document also includes the project schedule and budget, followed by appendices containing supporting information such as the glossary, abbreviations, references, and diagrams.

Chapter 2 — General Description

2.1 Product Perspective

SkillForge is an interactive puzzle-based skill development prototype designed as a self-contained project experience. The prototype combines an ancient-futuristic environment with interactive interface elements, progression systems, and puzzle-based challenges. Its experience is organized around a central HUB and connected challenge areas, allowing the user to progress through the defined prototype flow.

The prototype uses a fixed-camera, 2.5D presentation approach rather than a conventional free-roaming 3D gameplay model. The user interacts with the environment, interface panels, progression elements, and puzzle mechanisms through the defined interaction flow. The major progression path includes the opening experience, Hall Awakening, Guest HUB, account interaction, data feed, glyph conversion, Forge Vision progression, challenge interaction, portal activation, reward presentation, and return to the HUB.

SkillForge is being developed as a prototype for demonstration and evaluation.

The SRS therefore defines the requirements necessary for the approved prototype rather than describing a complete commercial-scale game.

The documented functionality will remain limited to the features finalized for the prototype.

2.2 Game Overview

SkillForge is an interactive puzzle-based skill development prototype set within an ancient-futuristic environment. The

experience begins with a cinematic opening sequence that introduces the world and leads the user into the Hall, followed by the Hall Awakening sequence and transition into the central HUB.

The HUB acts as the primary interaction and progression area of the prototype. From the HUB, the user can interact with the available interface elements, complete the account-related flow, receive the required data feed, and progress through the glyph and Forge systems. The progression ultimately enables access to the challenge experience. The challenge experience takes place within a dedicated challenge area where the user interacts with the defined puzzle mechanism. The challenge uses the Ring System, Alignment Analysis, Energy Flow, and Core Resonance as its major interaction stages. Successful completion activates the portal, presents the reward through the defined reward flow, and returns the user to the HUB.

The prototype therefore demonstrates a complete, controlled progression loop from the opening experience through

HUB interaction, progression, challenge completion, reward presentation, and return to the HUB. The experience is designed as a focused prototype rather than a large-scale free-roaming game.

2.3 Product Functions

Major Product Functions

Opening Experience — Provides the defined introductory sequence and transitions the user into the Hall and Hall Awakening experience.

HUB Management — Provides the central environment from which the user accesses the available progression and interaction systems.

Account Interaction — Provides the login/sign-up flow required for the user's progression.

Data Feed — Provides the required progression data after the account-related flow.

Glyph Conversion — Converts the relevant interface/progression elements into the defined glyph-based state.

Forge Progression — Enables the user to obtain and use the required Forge Vision progression.

Challenge Access — Provides access to the defined challenge experience after the required progression condition is satisfied.

Puzzle Interaction — Provides the Ring System, Alignment Analysis, Energy Flow, and Core Resonance stages required to complete the challenge.

Portal Activation — Activates the portal after successful challenge completion.

Reward Presentation — Presents the reward through the defined reward flow and completes the challenge progression.

Progress and Save Management — Maintains the defined challenge completion and progression behaviour.

Interface Support — Provides notifications, hints, settings, panels, and other required interface interactions.

Return to HUB — Returns the user to the central HUB after the defined challenge/reward flow.

2.4 User Community

2.4.1 Primary User — Player

The primary user of SkillForge is the player/user who interacts with the prototype. The player experiences the opening sequence and Hall Awakening, enters and interacts with the HUB, uses the available interface panels, performs the login/sign-up interaction, receives the data feed, progresses through the glyph and Forge systems, accesses the challenge experience, performs the puzzle interactions, activates the portal through successful challenge completion, receives the reward, and returns to the HUB.

2.4.2 User States

Guest User — The player before completing the account-related flow and before receiving the corresponding progression data.

Authenticated User — The player after completing the defined login/sign-up flow and receiving the required data feed, allowing progression through the subsequent systems.

Guest User and Authenticated User are not separate user communities or separate player roles. They represent two progression states of the same primary player.

No separate Admin, Teacher, Moderator, Multiplayer Player, or Content Creator roles are defined for the current prototype.

2.5 Operating Environment

2.5.1 Development Platform

SkillForge will be developed as a Windows desktop-based Unity prototype.

The project will use Unity 6.3 LTS as the game engine, C# as the primary programming language, Unity Hub for Unity installation and project/version management, a compatible C# development environment for script development, and

Windows 10/11 64-bit as the development operating system.

2.5.2 Hardware Requirements

Component

Recommended Development Configuration

Processor

AMD Ryzen 7 or equivalent modern multi-core processor

Memory

16 GB RAM or higher

Graphics

NVIDIA GeForce RTX 3050 4 GB/6 GB or better

Storage

SSD, 512 GB or higher

Operating System

Windows 10/11 64-bit

Display

Full HD or higher recommended

These are project-level recommended specifications, not Unity minimum official hardware requirements. Actual Editor performance depends on project complexity.

2.5.3 Target Platform

The current prototype will target Windows Desktop. The project is not currently specified for Android, iOS, console, WebGL, or other deployment platforms.

2.5.4 Project Environment

The approved project uses two Unity Scenes: Main Scene and Challenge Scene. The Main Scene contains the opening sequence, Hall, HUB, progression systems and related interface elements. The Challenge Scene contains the Challenge Chamber, Ring System, analysis systems, portal and reward transition.

Only entering a challenge causes a Unity Scene transition. Opening, HUB Awakening, Login, Data Feed and Glyph Conversion remain within the Main Scene.

The project follows a modular and reusable asset structure covering environment, gameplay, UI, effects and lighting prefabs.

2.5.5 Software and Runtime Support

The project will use Unity 6.3 LTS, C#, Unity UI systems, Unity audio systems, Unity scene management and Unity-supported Windows build tools.

For Windows development, the Unity environment must satisfy the supported operating-system and graphics requirements of the selected Unity LTS release.

2.6 Constraints

2.6.1 Development Constraints

SkillForge is being developed as a time-limited academic prototype. Therefore, development is restricted to the features necessary to demonstrate the approved gameplay flow and core systems.

The prototype will use a 2.5D, fixed-camera presentation rather than a conventional free-roaming 3D gameplay structure.

The project will target Windows Desktop for the current prototype.

2.6.2 Technical Constraints

The project architecture is based on two Unity Scenes only: MainScene and ChallengeScene.

Only challenge entry and return require scene transitions; UI-driven progression remains within the Main Scene. The architecture follows the rule: One Manager = One Responsibility. Managers must not duplicate ownership of systems or data.

Persistent data is handled only by SaveManager, while runtime player data belongs to PlayerManager.

2.6.3 Gameplay Constraints

The prototype is restricted to the approved progression flow.

The challenge system uses Forge Vision, Ring System, Alignment, Energy Flow, Core Resonance, Portal, and Reward flow.

A separate Reward Scene is not part of the architecture. The reward popup occurs within the portal transition.

2.6.4 Architecture Constraints

Scripts follow the single-responsibility principle.

Controllers communicate through Managers, while unnecessary Controller-to-Controller communication is avoided.

Save operations are handled only by SaveManager.

Reusable objects should use prefabs, while unique environment elements may remain scene objects.

The project follows a one-owner-per-data rule and avoids circular data flow and heavy event-bus architecture.

2.6.5 Documentation Constraints

The SRS will document only approved project requirements.

New gameplay or technical architecture decisions will not be introduced merely while writing the SRS.

The final prototype must be capable of demonstrating the functional requirements documented in the SRS.

2.7 Assumptions

2.7.1 Development Assumptions

The development team will have access to a compatible Windows development system capable of running the selected Unity LTS version.

Required Unity development tools and compatible C# scripting tools will be available during implementation.

The project team will have sufficient time to implement and test the approved prototype scope.

2.7.2 Technical Assumptions

Unity will provide the required scene, UI, animation, audio, lighting and rendering functionality needed by the approved prototype.

The selected development hardware will meet or exceed the recommended project configuration defined in Section 2.5.

Required project assets and supporting resources will be available or created during development.

2.7.3 Gameplay Assumptions

The player will follow the intended progression flow of the prototype.

The player will interact with the available UI and puzzle elements using the supported input method.

The challenge will be completed according to the defined puzzle interaction rules.

The prototype will be evaluated as an academic demonstration rather than as a complete commercial game.

2.7.4 Data and Persistence Assumptions

Local persistent storage will be available for the approved save functionality.

SaveManager will remain the sole persistent-data handler.

Player, progression and challenge data will follow the ownership rules defined in the approved Data Flow Architecture.

2.7.5 Documentation Assumptions

The approved technical architecture will remain the basis for implementation unless a deliberate architecture revision is made.

Features documented as functional requirements will be implemented sufficiently to demonstrate them in the prototype.

Chapter 3 — Functional Requirements

3.1 Opening Sequence

The Opening Sequence shall introduce the SkillForge world through a controlled cinematic sequence before transferring control to the player.

FR-3.1.1 — Game Launch

The system shall begin the SkillForge Opening Sequence when the application is launched. The initial presentation shall display the futuristic interface with the connection message and loading bar.

FR-3.1.2 — Fake Loading

The system shall display a simulated loading sequence during the opening. The loading progress shall visually advance through the predefined stages: 12% → 38% → 67% → 89% → 99%. The loading sequence shall intentionally stop at 99%.

FR-3.1.3 — 99% Freeze

The system shall keep the loading presentation frozen at 99% for a short period. The purpose of the freeze shall be to create the intended impression that the application has encountered a problem.

FR-3.1.4 — Glitch Transition

After the 99% freeze, the system shall initiate the predefined glitch transition. The loading bar shall flicker. A digital glitch sound shall be played. The background shall begin to visually distort.

FR-3.1.5 — Connection Failed

During the glitch transition, the displayed connection status shall change to: Connection Failed. The system shall then cut the screen to complete black.

FR-3.1.6 — Complete Darkness

After the connection failure, the system shall display a completely black screen. No user interface or loading indicator shall be displayed during this phase. The sequence shall temporarily remain silent before the ancient forge sound begins.

FR-3.1.7 — Forge Sound Sequence

The system shall play the predefined distant ancient forge-hammer sounds while the screen remains black. The sequence shall contain two distinct forge strikes separated by silence.

FR-3.1.8 — Connection Established

The system shall continue the opening sequence with the Connection Established with The SkillForge transition. This connection-established message is part of the approved final opening/transition flow and shall not be treated as a separate scene or separate sequence.

FR-3.1.9 — Blue Crystal Reveal

The system shall reveal a small glowing blue crystal at the centre of the dark screen. Blue particles shall surround the crystal and the defined humming sound shall accompany the reveal.

FR-3.1.10 — Ancient Narrative

The system shall present the approved narrative text using a typewriter-style presentation.

The narrative shall appear progressively and use the approved sequence of lines:

“Some places are built...

Some places are discovered.....

And

Some Places just exist !”

The text shall use the defined fade/pause transitions.

FR-3.1.11 — Crystal Awakening

The system shall initiate the Crystal Awakening sequence following the narrative. The sequence shall include: a powerful forge strike; increased crystal brightness; ancient orange energy; activation of circular patterns; expansion of a large energy wave; and transition to a completely white screen.

FR-3.1.12 — SkillForge Logo Reveal

The system shall stabilize the white screen and reveal the SkillForge logo. The logo shall remain visible briefly before fading into the following Hall reveal.

FR-3.1.13 — Hall Reveal

The system shall reveal the SkillForge Hall after the logo sequence. The reveal shall transition from the white light into the Hall environment. The Hall shall contain the approved ancient-futuristic environment elements, including the central crystal, Ancient Forge, Challenge Portal, floating environment, torches, mist and energy elements.

FR-3.1.14 — ENTER Prompt

After the Hall has been revealed, the system shall display the interaction prompt:

“Press ENTER to Begin.”

The player shall be required to activate the ENTER interaction to continue.

FR-3.1.15 — Hall / HUB Awakening

When the player activates ENTER, the system shall initiate the Hall Awakening sequence. The awakening shall include:

Ancient Forge activation; energy travelling from the Forge to the Crystal; Crystal absorbing the energy; energy pulse through the Hall; floor energy activation; Hall light activation; wall torch activation; holographic system activation; and floating environment activation. The sequence shall result in the awakened HUB environment.

FR-3.1.16 — Guest HUB

After the Hall Awakening sequence, the system shall enter the Guest HUB state. The major HUB panels shall be available in their Guest Mode without player progression data. Only the Profile Panel shall provide the Login/Sign-Up interaction at this stage. The player shall then proceed to the Login/Sign-Up stage to initialize the account and continue the approved progression flow.

3.1.17 — Opening Sequence: Complete Flow

Game Launch → Fake Loading → 12% → 38% → 67% → 89% → 99% → 99% Freeze → Glitch / Flicker / Distortion → Connection Failed → Black Screen → Forge Hammer Sounds → Connection Established with The SkillForge → Blue Crystal Reveal → Ancient Narrative → Crystal Awakening → SkillForge Logo Reveal → Hall Reveal → Press ENTER to Begin → Hall / HUB Awakening → Guest HUB.

3.2 Guest HUB

The Guest HUB shall be the first interactive HUB state after the completion of the Hall Awakening sequence and before the player creates or logs into an account.

FR-3.2.1 — Enter Guest HUB

After the Hall Awakening sequence is completed, the system shall enter the Guest HUB State. The Guest HUB shall remain within the Main Scene.

FR-3.2.2 — Guest HUB Presentation

The system shall display the major HUB panels in their Guest Mode. The panels shall be visible without player-specific gameplay or progression data. The Guest HUB shall present the approved futuristic/ancient holographic interface while keeping the HUB available for the subsequent account flow.

FR-3.2.3 — Guest Panel Interaction

During the Guest HUB state, the Profile Panel shall be the only major HUB panel that provides an account-related interaction. The player shall use the Profile Panel to initiate either:

- Login
- Sign-Up

Other HUB panels shall not provide normal progression interactions before account initialization.

FR-3.2.4 — Portal Restriction

The Challenge Portal shall not be used as the player's progression entry point while the player remains in the Guest HUB state. The player must first complete the Login/Sign-Up flow.

FR-3.2.5 — Guest HUB Panel Behaviour

The HUB shall follow the approved panel interaction model. Information panels use the defined Glyph → Expanded Panel → Back/ESC → Glyph behaviour. Special popup interfaces such as Login and Sign-Up follow popup behaviour and temporarily block background interaction while active.

FR-3.2.6 — Guest HUB Exit Behaviour

When the player presses ESC while in the Guest HUB, the system shall display the defined Exit Confirmation dialog.

The player shall be able to:

- Select Yes to close the game.
- Select No to return to the previous state.

FR-3.2.7 — Transition to Account Flow

When the player selects Login or Sign-Up from the Profile Panel, the system shall open the corresponding account popup. After successful account creation/login, the system shall generate/load the player's default account data and transition the HUB from Guest Mode toward the Active HUB state.

3.2.8 — Guest HUB Flow

Hall / HUB Awakening

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Guest HUB

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Major HUB Panels Visible Without Player Data

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Profile Panel

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Login / Sign-Up

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Account Initialization

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Default Player Data

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Active HUB

3.3 Login / Sign-Up

The Login / Sign-Up system shall allow the player to transition from the Guest HUB state into an initialized player account and Active HUB state.

FR-3.3.1 — Open Account Popup

The system shall provide Login and Sign-Up through the Profile Panel while the player is in the Guest HUB. Login and Sign-Up shall be presented as dedicated popup interfaces, rather than normal expandable HUB information panels.

FR-3.3.2 — Login Interface

When the player selects Login, the system shall display the Login popup. The Login popup shall temporarily block interaction with the underlying HUB until the popup is completed or closed.

FR-3.3.3 — Sign-Up Interface

When the player selects Sign-Up, the system shall display the Sign-Up popup. The Sign-Up popup shall follow the same dedicated-popup behaviour and shall temporarily block background HUB interaction while active.

FR-3.3.4 — Account Initialization

After successful account creation or login, the system shall initialize or load the player's account data. The approved transition flow specifies that default player data is generated/loaded after account creation or login.

FR-3.3.5 — Data Distribution to HUB

After account initialization, the system shall provide the generated/loaded player data to the relevant HUB panels. The panels shall transition from their Guest presentation toward the initialized Active HUB state.

FR-3.3.6 — Glyph Conversion

After the account data has been initialized, the major information panels shall convert into their defined Glyph Form. This establishes the normal Active HUB presentation. The normal HUB interaction model shall then follow: Glyph

→

Expanded Panel → Back / ESC → Glyph.

FR-3.3.7 — Login Success

After successful authentication, the system shall transition the player from Guest HUB toward the Active HUB state. A Login Successful notification shall be eligible to appear as a system notification.

FR-3.3.8 — Account Created

After successful account creation, the system shall generate the player's initial/default data and transition toward the Active HUB. An Account Created notification shall be eligible to appear as a system notification.

FR-3.3.9 — Login Failure

If the login attempt is unsuccessful, the system shall remain within the account flow and shall not transition the player into the Active HUB. A Login Failed notification is defined as an error notification.

FR-3.3.10 — Popup Behaviour

Login and Sign-Up popups shall follow the locked popup interaction rules:

- Background interaction is blocked while the popup is active.
- The popup remains above normal HUB panels.
- Popup interaction is completed or closed before returning to the underlying HUB state.

FR-3.3.11 — Transition to Active HUB

After successful Login/Sign-Up and account initialization, the system shall enter the Active HUB State. The Active HUB shall then provide the player's initialized progression data and enable the first approved objective: Forge Core → Collect Forge Vision → Skill Tree Update → Forge Core returns to Glyph.

3.3.12 — Login / Sign-Up Flow

Guest HUB

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Profile Panel

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Login / Sign-Up Popup

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Authentication / Account Creation

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Success

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Default Player Data Generated / Loaded

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HUB Panels Receive Data

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Panels Convert to Glyph Form

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Active HUB

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Account Created / Login Successful Notification

3.4 Data Feed

The Data Feed system shall present the player's initialized SkillForge-related information after successful account initialization and support the transition into the normal Active HUB progression.

FR-3.4.1 — Data Feed Access

After successful Login/Sign-Up and account initialization, the system shall make the Data Feed information available through the relevant HUB interface. The Data Feed shall form part of the player's initialized Active HUB experience.

FR-3.4.2 — Data Synchronization

The system shall synchronize the generated or loaded player data with the relevant HUB panels after account initialization. The account data shall be distributed according to the approved data ownership and manager architecture.

FR-3.4.3 — Data Presentation

The Data Feed shall present the approved SkillForge information in the defined holographic interface style. The system shall avoid duplicating information across panels; each information panel shall retain its defined responsibility.

FR-3.4.4 — Transition to Glyph State

After the required account data has been initialized and presented, the relevant information panels shall return to their defined Glyph Form for normal HUB interaction.

FR-3.4.5 — Active HUB Continuation

Completion of the initial Data Feed stage shall allow the player to continue with the first approved objective:

Forge Core

→ Collect Forge Vision → Skill Tree Update → Forge Core returns to Glyph.

3.4.6 — Data Feed Flow

Successful Login / Sign-Up

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Account Initialization

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Player Data Generated / Loaded

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Relevant HUB Panels Receive Data

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Data Feed Presentation

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Panels Transition to Glyph Form

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Active HUB Progression

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First Objective

3.5 Glyph Conversion

The Glyph Conversion system shall transform the initialized HUB information panels into the Glyph-based interface used during normal Active HUB interaction.

FR-3.5.1 — Start Glyph Conversion

After successful account initialization and Data Feed presentation, the system shall initiate the approved Glyph Conversion sequence. Glyph Conversion shall occur within the Main Scene and shall not create a Unity Scene transition.

FR-3.5.2 — Panel to Glyph Transformation

The relevant information panels shall transition from their expanded/data presentation into their defined Glyph Form.

The final HUB state shall use Glyphs as the default presentation for information panels.

FR-3.5.3 — Glyph Position

Each converted Glyph shall remain associated with its defined HUB position. When a player later activates a Glyph, the corresponding panel shall expand from the same position as the Glyph.

FR-3.5.4 — Glyph Interaction

The player shall be able to activate a Glyph to open its corresponding information panel. The interaction shall use the defined: Glyph → Expanded Panel transition. The panel shall provide its full interactive information interface after expansion.

FR-3.5.5 — Panel Return to Glyph

An expanded information panel shall be able to return to Glyph Form through:

- Its individual Back button.
- The global ESC behaviour.

FR-3.5.6 — Conversion Animation

The Glyph Conversion and subsequent panel interactions shall follow the approved holographic visual style. For normal panel opening: Glow → Expand → Ready. For closing: Collapse → Glow Fade → Glyph. The animation shall be fast, smooth and responsive, with the documented approximate interaction duration of 0.25–0.35 seconds, subject to implementation adjustment.

FR-3.5.7 — Default Active HUB State

After Glyph Conversion is complete, the Active HUB shall present information panels primarily in Glyph Form. The HUB shall therefore remain visually clean and uncluttered during normal interaction.

FR-3.5.8 — Multiple Glyph / Panel Behaviour

Multiple information panels may be opened simultaneously. Each panel shall operate independently and shall expand at its own original HUB position.

FR-3.5.9 — Hover Behaviour

When the player hovers over a Glyph, the system shall provide a temporary visual glow increase. When the pointer leaves the Glyph, it shall return to its normal brightness. No Glyph shall remain permanently highlighted.

3.5.10 — Glyph Conversion Flow

Data Feed Presentation

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Information Panels

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Glyph Conversion

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Glyph Form

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Player Selects Glyph

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Panel Expands at Same Position

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Panel Interaction

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Back / ESC

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Glyph Form

3.6 Forge Core

The Forge Core shall function as a primary progression interface in the Active HUB and shall provide access to the player's Forge-related progression.

FR-3.6.1 — Forge Core Availability

After the player enters the Active HUB and completes the initial Glyph Conversion, the Forge Core shall be available as a primary HUB interaction. The Forge Core is part of the Main Scene and does not require a separate Unity Scene.

FR-3.6.2 — Forge Core Interaction

The player shall be able to interact with the Forge Core through the defined HUB interaction system. The Forge Core shall use the approved holographic/interactive presentation rather than introducing an independent screen or scene.

FR-3.6.3 — First Forge Objective

The first approved Active HUB progression objective shall be: Forge Core → Collect Forge Vision → Skill Tree Update → Forge Core returns to Glyph.

FR-3.6.4 — Forge Vision Acquisition

The Forge Core progression shall allow the player to obtain/collect Forge Vision as the first major progression ability.

Forge Vision shall subsequently be used during Challenge Chamber gameplay, where it activates the analysis panels.

FR-3.6.5 — Skill Tree Update

After Forge Vision is collected, the system shall update the player's Skill Tree progression according to the approved first-objective flow.

FR-3.6.6 — Return to Glyph

After the Forge Core objective is completed, the Forge Core interface shall return to its defined Glyph presentation as part of the normal Active HUB state.

FR-3.6.7 — Progress Persistence

The resulting progression state shall be handled through the approved player progression and save architecture.

SaveManager is the sole persistent save handler, while PlayerManager owns runtime player data and ProgressManager handles progression-related UI/display state.

3.6.8 — Forge Core Flow

Active HUB

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Forge Core

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Collect Forge Vision

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Skill Tree Update

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Forge Core Returns to Glyph

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Active HUB

3.7 Forge Vision

The Forge Vision system shall provide the player with an analysis mode for observing and understanding the current state of challenge-related objects without directly revealing the puzzle solution.

FR-3.7.1 — Forge Vision Availability

After the player obtains Forge Vision through the Forge Core progression, the ability shall become available for use during Challenge gameplay.

FR-3.7.2 — Forge Vision Activation

The player shall activate Forge Vision using the defined V input. When activated, the system shall perform the Forge Vision scan and display the approved analysis panels.

FR-3.7.3 — Analysis Mode

While Forge Vision is active, the system shall provide analysis information for the current challenge formation.

ForgeVisionManager shall control Forge Vision state, analysis mode, scan updates and analysis panels.

FR-3.7.4 — Formation Status

The Formation Status analysis panel shall provide: Formation Name, Formation Type, Formation Integrity, and Current State.

FR-3.7.5 — Alignment Analysis

The Alignment Analysis panel shall provide: Alignment Accuracy, Structural Offset, Alignment Stability, and Structural Response.

FR-3.7.6 — Energy Flow Analysis

The Energy Flow panel shall provide: Energy Sources, Energy Flow, Energy Leakage, and Channel Stability.

FR-3.7.7 — Core Resonance Analysis

The Core Resonance panel shall provide: Resonance Level, Core State, Connection Quality, and Resonance Stability.

FR-3.7.8 — No Solution Reveal

Forge Vision shall provide analysis only. The system shall not directly reveal the correct puzzle solution, ring rotations, energy-routing instructions or other solution-specific guidance.

FR-3.7.9 — Forge Vision Deactivation

The player shall be able to deactivate Forge Vision using the defined V input. A first-time hint shall explain the deactivation action after the player's first activation.

FR-3.7.10 — Analysis Panel Availability

The challenge analysis panels shall be available only while Forge Vision is active.

FR-3.7.11 — Real-Time Analysis

The Forge Vision system shall update its analysis information as the challenge state changes. Real-time gameplay values such as Alignment, Energy Flow and Core Resonance shall be presented through the analysis panels rather than notifications.

3.7.12 — Forge Vision Flow

Forge Vision Acquired → Enter Challenge → Press V → Forge Vision Scan → Analysis Panels Appear → Player Solves Formation Using Analysis → Formation State Updates → Press V to Deactivate

3.8 Skill Tree

The Skill Tree shall provide the player with a progression interface for viewing and managing the abilities unlocked.

FR-3.8.1 — Skill Tree Availability

The Skill Tree shall become available as part of the Active HUB progression after the player obtains Forge Vision.

FR-3.8.2 — Skill Tree Update

After Forge Vision is collected, the system shall update the player's Skill Tree according to the approved first progression objective: Forge Core → Collect Forge Vision → Skill Tree Update → Forge Core returns to Glyph.

FR-3.8.3 — Skill Tree Presentation

The Skill Tree shall be accessible through its defined HUB interface and shall use the approved holographic presentation. The Skill Tree shall function as an information/progression panel rather than creating a separate Unity Scene.

FR-3.8.4 — Ability Progression

The Skill Tree shall represent the player's unlocked or available abilities according to the player's progression state. The system shall prevent progression information from being duplicated across unrelated HUB panels.

FR-3.8.5 — Progression State

The Skill Tree shall reflect the player's current progression state. Progression data shall follow the approved ownership architecture, with runtime player data handled by PlayerManager and persistent progression handled through SaveManager.

FR-3.8.6 — Glyph Behaviour

When not actively being viewed, the Skill Tree shall use its defined Glyph Form within the HUB. Selecting the Glyph shall open the corresponding Skill Tree panel, and the panel shall be capable of returning to Glyph Form through the standard Back/ESC behaviour.

FR-3.8.7 — Persistence

Unlocked progression associated with the Skill Tree shall be included in the player's persistent progression data according to the approved Save System.

3.8.8 — Skill Tree Flow

Active HUB → Skill Tree Glyph → Open Skill Tree → View Progression / Abilities → Return / ESC → Skill Tree Glyph

3.9 Challenge Selection

The Challenge Selection system shall allow the player to select an available challenge from the Challenge Portal interface before entering the selected challenge level.

FR-3.9.1 — Challenge Portal Access

After completion of the approved initial HUB progression, the player shall be able to interact with the Challenge Portal from the Active HUB. The Challenge Portal shall initiate the Challenge Selection flow.

FR-3.9.2 — Challenge Selection Interface

The system shall display the available challenge selection interface when the player interacts with the Challenge Portal. Challenge Selection shall be handled as a UI state within the approved Main Scene flow rather than as an independent gameplay scene.

FR-3.9.3 — Challenge Selection to Level Selection

After the player selects a challenge, the same selection panel shall transform into the Level Selection state. A separate Challenge Selection scene shall not be required for this transition.

FR-3.9.4 — Level Selection

The system shall display the available level-selection options for the selected challenge. The player shall select the required level before entering the Challenge Chamber.

FR-3.9.5 — Challenge Entry

After a level is selected, the system shall initiate the approved Portal Warp transition. The transition shall use the defined 1-second Black Fade before the Challenge Chamber becomes active.

FR-3.9.6 — Challenge Selection State

The UI state sequence shall follow: Active HUB → Challenge Selection → Level Selection → Challenge Chamber.

3.9.7 — Challenge Selection Flow

Active HUB

↓

Challenge Portal

↓

Challenge Selection

↓

Select Challenge

↓

Same Panel → Level Selection

↓

Select Level

↓

Portal Warp

↓

1-second Black Fade

↓

Challenge Chamber

3.10 Challenge Chamber

The Challenge Chamber shall provide the primary gameplay environment in which the player analyses and solves the selected Logic Challenge.

FR-3.10.1 — Enter Challenge Chamber

After the player selects a challenge level, the system shall perform the approved Portal Warp and 1-second Black Fade, then enter the Challenge Chamber.

FR-3.10.2 — Fixed Camera

The Challenge Chamber shall use a fixed camera presentation. The player shall not be able to move, zoom, or rotate the camera. The chamber shall remain visually stable during puzzle interaction.

FR-3.10.3 — Initial Challenge State

When the Challenge Chamber is entered, the system shall display the selected challenge in its initial unsolved state.

The chamber shall contain: Broken Formation, Inactive Portal, Challenge environment, Settings access, and Forge Vision access.

FR-3.10.4 — Settings Access

The Settings icon shall remain available within the Challenge Chamber. On the first Challenge Chamber entry, the Settings panel shall automatically open. The player shall be able to close it using the defined X interaction, after which the Settings icon shall remain available.

FR-3.10.5 — Forge Vision Access

The player shall be able to activate Forge Vision using the defined V input. Forge Vision shall reveal the approved analysis panels required for challenge analysis.

FR-3.10.6 — Analysis Panels

While Forge Vision is active, the Challenge Chamber shall provide Formation Status, Alignment Analysis, Energy Flow, and Core Resonance. These panels shall provide analysis information without directly revealing the puzzle solution.

FR-3.10.7 — Ring Interaction

The player shall solve the Logic Challenge by interacting with the Ring System. The challenge shall use three rings:

• Middle Ring — Physical Alignment • Outer Ring — Energy Flow • Inner Ring — Core Synchronization The rings shall be solved through the approved interaction rules.

FR-3.10.8 — Challenge Completion

The Challenge Chamber shall reach the completed state when Alignment becomes stable, Energy Flow becomes stable, and Core Resonance becomes stable. After these conditions are satisfied, the Formation shall become Stable and the Portal shall become Activated.

FR-3.10.9 — Portal Interaction

After the challenge is completed, the Portal shall become interactive. The player shall click the Portal to initiate the completion/transition sequence. There shall be no separate Continue / Enter Portal button.

FR-3.10.10 — Return Flow

Portal interaction shall initiate the approved Portal Warp and Reward Popup flow. The Reward shall be presented as a popup during the transition rather than through a separate Reward Scene.

3.10.11 — Challenge Chamber Flow

Challenge Selection → Level Selection → Portal Warp → 1-second Black Fade → Challenge Chamber → Broken Formation → Forge Vision → Analysis Panels → Ring Interaction → Alignment Stable → Energy Flow Stable → Core Resonance Stable → Formation Stable → Portal Activated → Click Portal → Portal Warp / Reward Flow

Chapter 3 — Functional Requirements

3.11 Ring System

The Ring System shall provide the primary interactive puzzle mechanism within the Challenge Chamber.

FR-3.11.1 — Ring Formation

The Challenge Chamber shall contain three interconnected rings forming the Logic Challenge: I Middle Ring — Physical Alignment I Outer Ring — Energy Flow I Inner Ring — Core Synchronization The three rings shall work together as a single puzzle formation.

FR-3.11.2 — Initial Ring State

When a challenge begins, the ring formation shall be presented in a Broken Formation state. The rings shall require player interaction before the formation can become stable.

FR-3.11.3 — Middle Ring

The Middle Ring shall control Physical Alignment. The player shall rotate the Middle Ring to align the required physical structure. The alignment state shall contribute to the overall Formation Stability.

FR-3.11.4 — Outer Ring

The Outer Ring shall control Energy Flow. The player shall rotate the Outer Ring to establish the required energy connection/flow. Energy Flow stability shall contribute to the overall completion state.

FR-3.11.5 — Inner Ring

The Inner Ring shall control Core Synchronization. The player shall rotate the Inner Ring to establish the required synchronization with the challenge core. Core Resonance stability shall contribute to the overall completion state.

FR-3.11.6 — Ring Rotation

The player shall interact with the rings through the approved rotation mechanism. Ring interaction shall update the corresponding challenge state in real time.

FR-3.11.7 — Forge Vision Feedback

When Forge Vision is active, the player shall be able to observe the current analytical state of the ring formation.

Forge Vision shall provide analysis but shall not directly reveal the correct ring rotations or solution.

FR-3.11.8 — Formation Stability

The Ring System shall continuously evaluate the combined state of Alignment + Energy Flow + Core Resonance. When all required states become stable, the Formation also do transition to Stable.

FR-3.11.9 — Portal Activation

Once the Formation becomes Stable, the Challenge Portal shall become activated. The Portal activation shall indicate successful completion of the Ring System.

FR-3.11.10 — Completion Notification

After successful completion, the system shall generate the defined challenge-completion notification. The notification shall communicate that the challenge has been completed without replacing the Challenge Chamber completion state.

3.11.11 — Ring System Flow

Broken Formation → Middle Ring — Physical Alignment → Outer Ring — Energy Flow → Inner Ring — Core Synchronization → Alignment Stable → Energy Flow Stable → Core Resonance Stable → Formation Stable → Portal Activated

3.12 Alignment Analysis

The Alignment Analysis system shall provide the player with information about the physical alignment state of the Challenge Formation while Forge Vision is active.

FR-3.12.1 — Alignment Analysis Availability

The Alignment Analysis panel shall be available only while Forge Vision is active.

FR-3.12.2 — Alignment Information

The Alignment Analysis panel shall display: I Alignment Accuracy I Structural Offset I Alignment Stability I Structural Response

FR-3.12.3 — Real-Time Update

The Alignment Analysis shall reflect changes in the challenge state as the player interacts with the Ring System. The analysis shall update as the physical alignment changes.

FR-3.12.4 — Analysis Only

The Alignment Analysis shall provide information about the current alignment state. It shall not directly reveal the correct ring rotation or solution.

FR-3.12.5 — Stable Alignment

When the required physical alignment is achieved, the Alignment state shall become Stable. Stable Alignment shall contribute to the overall Formation Stability condition required for Challenge completion.

3.12.6 — Alignment Analysis Flow

Forge Vision ON → Alignment Analysis Panel → Alignment Accuracy / Structural Offset / Stability / Response → Player Adjusts Middle Ring → Analysis Updates → Alignment Stable

3.13 Energy Flow

The Energy Flow system shall provide the player with information about the current energy-routing state of the Challenge Formation while Forge Vision is active.

FR-3.13.1 — Energy Flow Availability

The Energy Flow analysis panel shall be available only while Forge Vision is active.

FR-3.13.2 — Energy Flow Information

The Energy Flow panel shall display: I Energy Sources I Energy Flow I Energy Leakage I Channel Stability

FR-3.13.3 — Real-Time Update

The Energy Flow analysis shall reflect changes in the challenge state as the player interacts with the Ring System. The displayed energy information shall update according to the current challenge state.

FR-3.13.4 — Analysis Only

The Energy Flow panel shall provide information about the current energy state. It shall not directly reveal the correct Outer Ring rotation or the complete solution.

FR-3.13.5 — Stable Energy Flow

When the required energy-routing state is achieved, the Energy Flow state shall become Stable. Stable Energy Flow shall contribute to the overall Formation Stability condition required for Challenge completion.

3.13.6 — Energy Flow Analysis

Forge Vision ON → Energy Flow Panel → Energy Sources / Flow / Leakage / Channel Stability → Player Adjusts Outer Ring → Energy Analysis Updates → Energy Flow Stable

3.14 Core Resonance

The Core Resonance system shall provide the player with information about the current synchronization state of the Challenge Formation while Forge Vision is active.

FR-3.14.1 — Core Resonance Availability

The Core Resonance analysis panel shall be available only while Forge Vision is active.

FR-3.14.2 — Core Resonance Information

The Core Resonance panel shall display: I Resonance Level I Core State I Connection Quality I Resonance Stability

FR-3.14.3 — Real-Time Update

The Core Resonance analysis shall reflect changes in the challenge state as the player interacts with the Ring System.

FR-3.14.4 — Analysis Only

The Core Resonance panel shall provide information about the current core synchronization state. It shall not directly reveal the correct Inner Ring position or complete puzzle solution.

FR-3.14.5 — Stable Core Resonance

When the required synchronization state is achieved, Core Resonance shall become Stable. Stable Core Resonance shall contribute to the overall Formation Stability condition required for Challenge completion.

3.14.6 — Core Resonance Flow

Forge Vision ON → Core Resonance Panel → Resonance Level / Core State / Connection Quality / Resonance Stability → Player Adjusts Inner Ring → Resonance Analysis Updates → Core Resonance Stable

3.15 Portal Ignition

The Portal Ignition system shall activate the Challenge Portal after the player successfully completes the Logic Challenge and shall provide the approved transition into the reward flow.

FR-3.15.1 — Formation Completion

The Logic Challenge shall be considered complete when the required challenge states become stable: Alignment, Energy Flow, and Core Resonance.

FR-3.15.2 — Formation Stable

After all required challenge states become stable, the Formation shall transition to the Stable state.

FR-3.15.3 — Portal Ignition

After the Formation becomes Stable, the Challenge Portal shall ignite/become activated and shall become available for player interaction.

FR-3.15.4 — Portal Interaction

The player shall click the activated Portal to initiate the approved completion transition. No separate Continue, Next, or Enter Portal button shall be required.

FR-3.15.5 — Portal Warp

After Portal interaction, the system shall initiate the approved Portal Warp transition.

FR-3.15.6 — Reward Transition

The Portal Warp shall lead to the defined Reward Popup/Reward Panel within the Portal transition. A separate Reward Chamber shall not be used.

FR-3.15.7 — First Challenge Completion Flow

For the first challenge, the approved flow shall be: Challenge Complete → Click Portal → Portal Warp → Reward Panel inside Portal Transition.

3.15.8 — Portal Ignition Flow

Alignment Stable + Energy Flow Stable + Core Resonance Stable

↓

Formation Stable

↓

Portal Ignites / Becomes Activated

↓

Player Clicks Portal

↓

Portal Warp

↓

Reward Popup / Reward Panel

3.16 Reward System

The Reward System shall present and apply the reward earned by the player after successful completion of a Challenge.

FR-3.16.1 — Reward Trigger

After the player successfully completes the Challenge and clicks the activated Portal, the system shall initiate the approved Portal Warp and Reward flow.

FR-3.16.2 — Reward Popup / Panel

The reward shall be presented through the defined Reward Popup/Reward Panel within the Portal transition. The Reward shall not be presented through a separate Reward Chamber.

FR-3.16.3 — Reward Collection

The player shall collect the displayed reward through the approved Reward interaction. The Reward Popup is a popup UI state and shall block background interaction while active.

FR-3.16.4 — Reward Application

After the reward is collected, the reward shall be applied to the player's progression according to the approved progression and save rules.

FR-3.16.5 — Return to HUB

After reward collection, the system shall return the player to the HUB through the approved Portal Warp and 1-second Black Fade transition.

FR-3.16.6 — Progress Update

After returning to the HUB, the relevant progression/panel information shall update and the Notification Queue shall receive the applicable notification.

FR-3.16.7 — Reward Flow

Challenge Complete → Click Portal → Portal Warp → Reward Popup / Reward Panel → Collect Reward → Reward Applied → Save Progress → Portal Warp Completes → 1-second Black Fade → HUB → Relevant Panels Update → Notification Queue

3.17 Save System

The Save System shall maintain the approved persistent player progression, completed Challenge state, rewards and relevant settings data for the prototype.

FR-3.17.1 — Persistent Save Ownership

SaveManager shall be the sole persistent save handler. Other systems may provide data to SaveManager, but persistent storage shall not be handled by other managers or controllers.

FR-3.17.2 — Completed Challenge Save

When a Challenge is completed, the system shall save the completed Challenge progress and associated reward state.

FR-3.17.3 — Reward Persistence

Rewards obtained from completed Challenges shall be saved as persistent player progression. Each completed Challenge shall grant its reward only once in the prototype.

FR-3.17.4 — Incomplete Challenge Exit

If the player exits a Challenge before completion, no incomplete Challenge progress or reward shall be saved. The Challenge shall start fresh on the next attempt.

FR-3.17.5 — Restart

Restart shall reset the Challenge to its original starting state, reset its analysis values, and grant no reward or Challenge progress for that restart.

FR-3.17.6 — Persistent Player Data

The Save System shall persist the approved player data required for the prototype, including progression, rewards/inventory and relevant settings data.

FR-3.17.7 — Save Data Sources

SaveManager shall receive the required persistent data from the approved data owners, including PlayerManager, SettingsManager and ChallengeManager. ProgressManager shall provide progression display/state and shall not perform persistent saving.

FR-3.17.8 — Auto-Save / Completion Flow

After completed Challenge reward processing, the updated runtime player/progression data shall be handled by PlayerManager and the required persistent state shall be passed to SaveManager before the player returns to the HUB.

3.17.9 — Save Flow

Completed Challenge → Rewards Applied → Progress Updated → SaveManager → Persistent Save → HUB Return

Incomplete Challenge → Exit / Restart → Nothing Saved → Challenge Starts Fresh

3.18 Notification System

The Notification System shall provide short, non-intrusive feedback for meaningful system, challenge, reward, skill and error events.

FR-3.18.1 — Notification Trigger

Notifications shall be generated only for meaningful events. Panel opening, panel closing, hover and similar routine interactions shall not generate notifications.

FR-3.18.2 — Notification Position

Notifications shall appear at the Top-Center of the screen.

FR-3.18.3 — Notification Animation

Notifications shall use the sequence: Fade In → Stay → Fade Out. The animation shall remain simple, clean and non-distracting.

FR-3.18.4 — Notification Duration

Normal notifications shall remain visible for approximately 2.5–3 seconds. Critical notifications may remain until the required interaction or duration.

FR-3.18.5 — Notification Categories

System: Account Created; Login Successful; Forge Vision Acquired; Settings Saved (Optional). Challenge:

Challenge

Completed. Reward: Reward Acquired. Skill: Skill Unlocked: [Skill Name]. Error: Invalid Input; Login Failed; Future system errors.

FR-3.18.6 — Notification Queue

When multiple notifications are generated, they shall appear one after another. Multiple notifications shall never appear simultaneously.

FR-3.18.7 — Gameplay Information Rule

Real-time gameplay values such as Alignment, Energy Flow and Core Resonance shall be displayed through the appropriate analysis panels and not as notifications.

FR-3.18.8 — Notification vs Popup

A Notification shall be small, temporary and automatically disappear. A Popup shall require player interaction, such as Login, Reward or Welcome.

FR-3.18.9 — Reward Notification

XP shall not generate a separate XP Added notification. XP information shall be included within Reward Acquired.

3.18.10 — Notification Flow

Game Event → NotificationManager → Notification Queue → Top-Center Notification UI → Fade In → Stay → Fade Out → Next Notification

3.19 Settings

The Settings system shall provide the player with access to the approved audio and interface settings used by the prototype.

FR-3.19.1 — Settings Access

The player shall be able to open the Settings panel through the available Settings interface. In the Challenge Chamber, the Settings icon shall remain available at the top-right.

FR-3.19.2 — Settings Ownership

Settings logic shall be handled by SettingsManager. SettingsManager shall handle audio settings, future graphics/gameplay settings, applying settings and save requests.

FR-3.19.3 — Audio Settings

The Settings system shall provide the approved audio controls for the prototype. Audio settings shall communicate with AudioManager for application of the relevant audio changes.

FR-3.19.4 — Settings Apply

When the player applies a setting, the system shall apply the selected value through SettingsManager.

FR-3.19.5 — Settings Persistence

Approved persistent settings shall be passed to SaveManager for saving. The approved data flow is: UIManager → SettingsManager → Apply Settings → AudioManager → SaveManager.

FR-3.19.6 — Challenge Chamber Settings Behaviour

On the first Challenge Chamber entry, the Settings panel shall open automatically. The player shall close it using X.

Afterward, only the Settings icon shall remain visible, and the panel shall open again only when the player clicks the icon.

FR-3.19.7 — Settings and ESC

Settings shall follow the approved UI state and ESC behaviour. ESC shall not create an invalid UI state or bypass the defined popup/panel navigation rules.

3.19.8 — Settings Flow

Settings Icon → Settings Panel → Change Setting → Apply → SettingsManager → AudioManager / Relevant System → SaveManager

3.20 Return to HUB

The Return to HUB system shall restore the player to the Main Scene after the approved Challenge completion, reward or manual-exit flow.

FR-3.20.1 — Completed Challenge Return

After the player collects the Reward, the system shall initiate the approved return-to-HUB sequence: Collect Reward & Reward Applied → Save Progress → Portal Warp Completes → 1-second Black Fade → HUB → Progress Updates.

FR-3.20.2 — Main Scene Restoration

The player shall return from the Challenge Scene to the Main Scene. The Main Scene shall restore the HUB with the updated saved player data.

FR-3.20.3 — Progress Update

After returning to the HUB, the relevant progression information shall update. The approved Reward flow shall pass through ChallengeManager → PlayerManager → SaveManager → NotificationManager → UIManager → Return HUB, with ProgressManager updating the related progression display/state.

FR-3.20.4 — Notification Queue

After the reward/progression update, applicable notifications shall be sent to the Notification Queue.

NotificationManager shall manage the queue before the notification is displayed through the UI.

FR-3.20.5 — Manual Challenge Exit

If the player exits a Challenge before completion, the system shall follow: Back → Leave Challenge? → Yes → Portal Warp → 1-second Black Fade → HUB. No reward or incomplete Challenge progress shall be saved.

FR-3.20.6 — No Invalid State

The Return to HUB flow shall follow the approved UI and Scene state transitions. Invalid state jumps shall not be permitted.

3.20.7 — Return Flow

Challenge Scene → Portal Warp → 1-second Black Fade → Main Scene → HUB → Progress Updates → Notification Queue

Chapter 4 — User Interface Requirements

4.1 HUB Interface

The HUB Interface shall provide the primary interactive interface through which the player accesses the approved progression, information and interaction systems of the SkillForge prototype.

UI-4.1.1 — HUB Presentation

The HUB shall use the approved futuristic/ancient holographic interface presentation within the fixed-camera 2.5D environment. The HUB shall remain part of the Main Scene and shall provide access to the defined HUB interaction elements.

UI-4.1.2 — HUB States

The HUB interface shall support the approved progression states: Guest HUB and Active HUB. Guest HUB shall be used before account initialization, while Active HUB shall be used after successful Login/Sign-Up and account initialization.

UI-4.1.3 — HUB Panels

The HUB shall provide the approved interface panels and interactive elements required by the prototype, including the relevant Profile, Forge, Skill Tree, Challenge, Knowledge/Information and Settings interfaces according to the player's current progression state.

UI-4.1.4 — Guest HUB Restrictions

During Guest HUB state, the Profile Panel shall provide the account-related interaction required to access Login and Sign-Up. Other progression panels shall not provide normal progression interactions before account initialization.

UI-4.1.5 — Active HUB

After successful account initialization, the interface shall transition from Guest HUB toward Active HUB. The Active HUB shall provide the player's available progression and interaction panels according to the current progression state.

UI-4.1.6 — Panel Interaction

Information panels shall follow the approved Glyph → Expanded Panel → Back/ESC → Glyph interaction model.

Popup interfaces shall follow popup-state behaviour and shall block background interaction while active.

UI-4.1.7 — ESC Behaviour

In the Guest HUB, pressing ESC shall open the defined Exit Confirmation. In the Active HUB, pressing ESC shall close open panels back to their Glyph states. Pressing ESC again shall open Exit Confirmation.

UI-4.1.8 — HUB Progression Access

The HUB shall provide access to the approved progression flow without introducing additional gameplay systems outside the prototype scope. The Challenge Portal shall become a valid progression entry point only after the required progression/account conditions have been satisfied.

4.1.9 — HUB Interface Flow

Opening / Hall Awakening → Guest HUB → Profile → Login / Sign-Up → Account Initialization → Active HUB → Available HUB Panels / Progression

4.2 Challenge Interface

The Challenge Interface shall provide the complete user interface required for interacting with the approved Challenge Chamber and its puzzle progression.

UI-4.2.1 — Challenge Presentation

The Challenge Interface shall present the Challenge Chamber within the approved fixed-camera 2.5D environment.

UI-4.2.2 — Challenge UI Elements

The interface shall provide the approved Challenge controls and information elements required for the Challenge, including Challenge identification, Challenge progress information, Ring System interaction area, Forge Vision/analysis interface, Settings access, Back/Challenge Exit control, and relevant analysis panels.

UI-4.2.3 — Ring Interaction

The Challenge Interface shall provide the interaction area required for the Ring System. The player shall be able to interact with the Ring System according to the approved puzzle interaction requirements.

UI-4.2.4 — Analysis Interface

The interface shall provide access to the approved analysis information for Alignment, Energy Flow and Core Resonance. These values shall be presented through their respective analysis panels rather than through notifications.

UI-4.2.5 — Settings Access

The Settings icon shall remain available at the approved top-right position during the Challenge Chamber. On the first Challenge Chamber entry, the Settings panel shall open automatically. After the player closes it, the Settings icon shall remain available for subsequent access.

UI-4.2.6 — Challenge Exit

The Challenge Interface shall provide the approved Back/exit interaction. Selecting Back shall display the Leave Challenge? confirmation. Yes → Portal Warp → 1-second Black Fade → HUB. No → Continue Challenge. An incomplete Challenge exit shall not grant a reward.

UI-4.2.7 — Completion Transition

After successful Challenge completion, the Challenge Interface shall transition into the approved Portal and Reward

flow: Challenge Complete → Click Portal → Portal Warp Begins → Reward Popup. The Reward shall remain a Popup State and shall not become a separate world/interface state.

4.2.8 — Challenge Interface Flow

Challenge Selection → Challenge Chamber → Ring System → Alignment / Energy Flow / Core Resonance →
Challenge Complete → Click Portal → Portal Warp Begins → Reward Popup

4.3 Analysis Panels

The Analysis Panels shall provide the player with non-solution-revealing information about the current state of the Challenge while Forge Vision is active.

UI-4.3.1 — Analysis Panel Access

The Analysis Panels shall be accessible through Forge Vision. The approved analysis panels shall include Formation Status, Alignment Analysis, Energy Flow and Core Resonance.

UI-4.3.2 — Formation Status

The Formation Status panel shall communicate the current overall state of the Challenge Formation and support the player in understanding whether the Formation is progressing toward the required Stable state.

UI-4.3.3 — Alignment Analysis Panel

The Alignment Analysis panel shall provide: Alignment Accuracy, Structural Offset, Alignment Stability and Structural Response.

UI-4.3.4 — Energy Flow Panel

The Energy Flow panel shall provide: Energy Sources, Energy Flow, Energy Leakage and Channel Stability.

UI-4.3.5 — Core Resonance Panel

The Core Resonance panel shall provide: Resonance Level, Core State, Connection Quality and Resonance Stability.

UI-4.3.6 — Analysis Behaviour

The analysis panels shall update according to the current Challenge state while the player interacts with the Ring System. The panels shall provide analysis information only and shall not directly reveal the correct solution.

UI-4.3.7 — Analysis vs Notification

Alignment, Energy Flow and Core Resonance values shall be displayed through the Analysis Panels and shall not be displayed as temporary notifications.

UI-4.3.8 — Panel Interaction

Analysis panels shall follow the approved panel interaction model. The player shall be able to open and close the relevant analysis information without leaving the Challenge Chamber.

4.3.9 — Analysis Panel Flow

Forge Vision → Formation Status → Alignment Analysis → Energy Flow → Core Resonance → Player Analyses
Current State

4.4 Reward Popup

The Reward Popup shall present the reward obtained after successful Challenge completion as a popup UI state during the approved Portal transition.

UI-4.4.1 — Reward Popup State

Reward shall be implemented as a Popup State. While the Reward Popup is active, background interaction shall be blocked.

UI-4.4.2 — Reward Flow

The approved flow shall be: Challenge Complete → Click Portal → Portal Warp Begins → Reward Popup → Collect Reward
→ Reward Applied → Save Progress → Portal Warp Completes → 1-second Black Fade → HUB → Progress Updates.

UI-4.4.3 — Reward Interaction

The Reward Popup shall provide the approved interaction for collecting the displayed reward and returning to the HUB. The Reward Popup shall not require a separate Reward Chamber or Reward Scene.

UI-4.4.4 — Reward Information

The popup shall present the reward information defined by the completed Challenge. XP shall be included within the
Reward Acquired presentation and shall not generate a separate XP Added notification.

UI-4.4.5 — Popup Exit Behaviour

ESC shall be disabled while the Reward Popup is active. The player shall use the approved Reward interaction to proceed.

UI-4.4.6 — Reward vs Notification

The Reward Popup shall remain distinct from temporary Notifications. A Popup requires player interaction, while a Notification is small, temporary and automatically disappears.

4.4.7 — Reward Popup Flow

Challenge Complete → Click Portal → Portal Warp Begins → Reward Popup → Collect Reward → Reward Applied →
Save Progress → Portal Warp → 1-second Black Fade → HUB → Progress Updates

4.5 Notification System

The Notification System shall provide short, non-intrusive feedback for meaningful system, challenge, reward, skill and error events.

UI-4.5.1 — Notification Position

Notifications shall appear at the Top-Center of the screen.

UI-4.5.2 — Notification Animation

Notifications shall use Fade In → Stay → Fade Out. The animation shall remain simple, clean and non-distracting.

UI-4.5.3 — Notification Duration

Normal notifications shall remain visible for approximately 2.5–3 seconds. Critical notifications may remain until the required interaction or duration.

UI-4.5.4 — Notification Categories

System: Account Created; Login Successful; Forge Vision Acquired; Settings Saved (Optional).

Challenge: Challenge Completed.

Reward: Reward Acquired.

Skill: Skill Unlocked: [Skill Name].

Error: Invalid Input; Login Failed; Future system errors.

UI-4.5.5 — Queue Behaviour

Notifications shall appear one after another. Multiple notifications shall never appear simultaneously.

UI-4.5.6 — Gameplay Information

Real-time values for Alignment, Energy Flow and Core Resonance shall remain within their Analysis Panels and shall not be displayed as Notifications.

UI-4.5.7 — Notification vs Popup

A Notification shall be small, temporary and automatically disappearing. A Popup shall require player interaction.

UI-4.5.8 — Reward Notification

There shall be no separate XP Added notification. XP shall be included within Reward Acquired.

4.5.9 — Notification Flow

Meaningful Event → Notification → Top-Center → Fade In → Stay → Fade Out → Next Notification

4.6 Hint System

The Hint System shall guide the player through required interactions without revealing the solution to the Challenge.

UI-4.6.1 — Hint Philosophy

Hints shall guide the player and shall not solve the puzzle.

UI-4.6.2 — Allowed Hints

The approved hints shall include: Press ENTER; Press V to Activate Forge Vision; Press V to Deactivate Forge Vision (first-time only); Click the Portal (first completed Challenge only).

UI-4.6.3 — Prohibited Hints

The system shall not provide gameplay-solution hints such as Rotate Left, Rotate Middle Ring, Try Clockwise, or specific energy-routing instructions.

UI-4.6.4 — Hint Position & Style

Hints shall appear at the Bottom-Center of the screen and use a semi-transparent, minimal, holographic presentation.

UI-4.6.5 — Hint Persistence

Hints shall not disappear automatically. Each hint shall remain visible until the required player action is successfully completed.
After completion, that hint shall be permanently removed.

UI-4.6.6 — One-Hint Rule

Only one hint shall be displayed at a time. The next hint shall appear only after the previous hint has been completed.

UI-4.6.7 — Forge Vision Hints

Forge Vision shall be mandatory for solving Challenges. Analysis Panels shall be available only while Forge Vision is active. After the first Forge Vision activation, a first-time hint shall explain how to deactivate Forge Vision using V.

UI-4.6.8 — Reward Hint

No separate Click Reward hint shall be displayed. The Reward Panel's own Collect Reward & Return to HUB interaction shall guide the player.

UI-4.6.9 — Portal Hint

After the first completed Challenge only, the system shall display: Click the Portal. After the player clicks the Portal once, this hint shall never appear again.

4.6.10 — Hint Flow

Required Player Action → Relevant Hint Appears → Player Performs Action → Hint Removed Permanently → Next Hint, If Applicable

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4.7 Settings Panel

The Settings Panel shall provide the player's approved settings controls while maintaining the defined UI state behaviour.

UI-4.7.1 — Settings Panel Access

The Settings Panel shall be accessible through the approved Settings icon. During the Challenge Chamber, the Settings icon shall remain available at the top-right.

UI-4.7.2 — First Challenge Entry

On the first Challenge Chamber entry, the Settings Panel shall open automatically. The player shall close it using X.

Afterward, the Settings Panel shall remain closed until the player selects the Settings icon.

UI-4.7.3 — Settings State

The Settings Panel shall follow the approved panel/popup state behaviour. When the panel is active, the player shall be able to interact with the available settings controls without creating an invalid UI state.

UI-4.7.4 — Audio Controls

The Settings Panel shall provide the approved audio controls for the prototype. Changes shall be applied through the defined Settings system and Audio system.

UI-4.7.5 — Settings Persistence

Approved persistent settings shall be passed to the Save System according to the defined data flow.

UI-4.7.6 — Close Behaviour

The player shall be able to close the Settings Panel using the approved X control. The panel shall return to its previous UI state without changing the underlying gameplay state.

4.7.7 — Settings Flow

Settings Icon → Settings Panel → Change Setting → Apply → Close / X → Previous UI State

4.8 Screen Flow

The Screen Flow shall define the approved navigation and transition sequence between the major screens and states of the SkillForge prototype.

UI-4.8.1 — Opening Flow

Opening Sequence → Guest HUB. The approved Opening Sequence shall transition into the Guest HUB after the opening cinematic/initialization sequence.

UI-4.8.2 — Account Flow

Guest HUB → Profile → Login / Sign-Up → Account Initialization → Active HUB. The account flow shall follow the approved Guest-to-Active HUB transition.

UI-4.8.3 — Challenge Flow

Active HUB → Challenge Selection → Challenge Chamber. The player shall enter the selected Challenge through the approved Challenge Selection interface.

UI-4.8.4 — Challenge Completion Flow

Challenge Chamber → Formation Stable → Portal Ignition → Portal Click → Portal Warp → Reward Popup. The Reward Popup shall remain part of the approved Portal transition.

UI-4.8.5 — Return Flow

Reward Collection → Reward Applied → 1-second Black Fade → HUB → Progress Update.

UI-4.8.6 — Manual Challenge Exit

For an incomplete Challenge: Back → Leave Challenge? → Yes → Portal Warp → 1-second Black Fade → HUB.

Selecting No shall return the player to the Challenge.

UI-4.8.7 — Panel Navigation

Information Panel: Glyph → Expanded Panel → Back / ESC → Glyph. Popup: Trigger → Popup Active → Required Interaction → Popup Closed.

UI-4.8.8 — Invalid State Prevention

Screen transitions shall follow the approved UI State Machine. The system shall not allow invalid jumps between unrelated UI states or scenes.

4.8.9 — Overall Screen Flow

Opening → Guest HUB → Login / Sign-Up → Active HUB → Challenge Selection → Challenge Chamber → Portal Ignition → Reward Popup → HUB

4.9 Panel Behaviour

The Panel Behaviour requirements shall define how approved panels, popups and information interfaces open, close and interact with the underlying UI state.

UI-4.9.1 — Information Panel Behaviour

Information panels shall follow: Glyph → Expanded Panel → Back / ESC → Glyph. Opening an information panel shall not create a new world or scene state.

UI-4.9.2 — Popup Behaviour

Popups shall follow: Trigger → Popup Active → Required Interaction → Popup Closed. While a popup is active, background interaction shall be blocked.

UI-4.9.3 — Notification Behaviour

Notifications shall remain separate from popup states. Notifications are temporary and automatically disappear, while popups require player interaction.

UI-4.9.4 — Settings Panel Behaviour

The Settings Panel shall follow the approved Settings behaviour. On first Challenge Chamber entry it opens automatically; after closing, it shall open again only through the Settings icon.

UI-4.9.5 — Reward Popup Behaviour

The Reward Popup shall remain a Popup State and shall block background interaction until the required reward interaction is completed.

UI-4.9.6 — ESC Behaviour

ESC shall follow the approved state-dependent behaviour: Open information panel → ESC closes it; Active HUB with no open panel → ESC opens Exit Confirmation; Guest HUB → ESC opens Exit Confirmation; Reward Popup → ESC disabled.

UI-4.9.7 — State Consistency

The UI shall not permit invalid state transitions or interaction with background elements that are blocked by the active panel/popup state.

4.9.8 — Panel Behaviour Flow

Trigger → Panel / Popup Active → Background Interaction Restricted if Required → Approved Interaction → Panel / Popup Closed → Previous Valid UI State

4.10 Software Interfaces

The Software Interfaces section shall define the external and internal software interfaces required for the SkillForge prototype at the SRS level.

UI-4.10.1 — Unity Engine

The prototype shall be developed using the approved Unity Engine environment. Unity shall provide the runtime environment for the game's scenes, UI, player interaction, challenge systems and visual presentation.

UI-4.10.2 — Input Interface

The prototype shall use the approved keyboard/mouse input interactions required by the gameplay and UI specifications, including the defined interactions for navigation, panel interaction, Ring System interaction, Forge Vision, Back/Exit and Reward interaction.

UI-4.10.3 — Audio Interface

The Audio system shall provide the interface required for applying the approved audio settings and playing the prototype's defined audio feedback. The Settings system shall communicate with the Audio system through the approved manager architecture.

UI-4.10.4 — Save Interface

The UI and gameplay systems shall communicate persistent player data through SaveManager. SaveManager shall provide the approved persistent storage interface for progression, rewards and other defined persistent data.

UI-4.10.5 — Manager Interfaces

The prototype's approved systems shall communicate through the defined manager architecture, including PlayerManager, ProgressManager, ChallengeManager, SettingsManager, SaveManager, NotificationManager, UIManager and AudioManager.

UI-4.10.6 — Scene Interface

The prototype shall use the approved scene structure for the Main Scene / HUB and Challenge Scene. Scene transitions shall follow the approved Screen Transition and UI State Machine specifications.

UI-4.10.7 — Interface Constraint

This section shall define software interfaces at the requirements/architecture level only. Specific class implementations, APIs, code structure and low-level programming details shall be defined during implementation and shall not be unnecessarily specified in the SRS.

4.10.8 — Software Interface Flow

UI / Gameplay Event → Relevant Manager → Required System → Save / Audio / Notification / Scene Interface

Chapter 5 — Performance Requirements

5.1 Response Time

The SkillForge prototype shall provide responsive interaction for the approved gameplay and UI operations.

PR-5.1.1 — User Interaction Response

The prototype shall respond to normal user interactions without noticeable or unnecessary delay. This includes UI button/panel interaction, Ring System interaction, Forge Vision activation/deactivation, Challenge interaction, Settings interaction and Reward interaction.

PR-5.1.2 — UI Response

When the player interacts with an available UI control, the corresponding approved UI state or action shall respond promptly and shall not produce unnecessary waiting.

PR-5.1.3 — Gameplay Response

Gameplay interactions within the Challenge Chamber shall respond without noticeable input lag under the intended prototype operating environment.

PR-5.1.4 — Transition Response

Approved screen and scene transitions shall begin promptly after their triggering interaction. The transition shall follow the defined transition behaviour rather than introducing an unplanned waiting state.

PR-5.1.5 — Prototype Constraint

Response-time requirements shall be evaluated under the intended prototype environment and shall not be interpreted as commercial-scale performance guarantees.

5.1.6 — Response Flow

Player Input → System Processes Input → Approved UI / Gameplay Response

5.2 Smooth Transitions

The prototype shall provide smooth visual transitions between approved UI states, scenes and gameplay states.

PR-5.2.1 — Transition Smoothness

Approved transitions shall appear smooth and visually continuous without unnecessary interruption or abrupt state changes.

PR-5.2.2 — Screen Transitions

Transitions between the Main Scene/HUB and Challenge Scene shall follow the approved transition behaviour.

PR-5.2.3 — Black Fade

Where the approved flow specifies a 1-second Black Fade, the transition shall use that defined duration. This applies to the approved Portal Warp / HUB return flow.

PR-5.2.4 — Portal Transition

The Portal transition shall maintain the approved sequence: Portal Click → Portal Warp → Reward Popup. The transition shall not introduce an additional intermediate scene or Reward Chamber.

PR-5.2.5 — UI State Transitions

Panel and popup transitions shall follow their approved UI state changes without visual or interaction conflicts.

PR-5.2.6 — Transition Consistency

The same approved transition behaviour shall be maintained wherever the corresponding transition is used in the SkillForge.

5.2.7 — Transition Flow

Trigger → Approved Transition → Target UI / Scene State

5.3 Animation Performance

The prototype shall maintain acceptable animation performance for the approved visual effects and transitions.

PR-5.3.1 — Animation Responsiveness

Approved animations shall play responsively without causing noticeable interruption to normal player interaction.

PR-5.3.2 — UI Animations

UI animations such as Panel transitions, Popup transitions, Notification Fade In / Stay / Fade Out and Hint presentation shall remain visually smooth and shall not interfere with required interaction.

PR-5.3.3 — Gameplay Animations

Gameplay-related visual effects, including the approved Portal, Forge Vision and Challenge visual feedback, shall operate without causing noticeable degradation of the prototype's interactive experience.

PR-5.3.4 — Transition Effects

The approved 1-second Black Fade shall maintain its defined duration where specified. Other animation durations shall follow their approved UI/visual specifications.

PR-5.3.5 — Prototype-Level Constraint

Animation performance shall be evaluated within the intended prototype environment. The prototype shall not be required to meet commercial-scale animation-performance targets.

5.3.6 — Animation Performance Flow

Animation Trigger → Animation Plays → Player Interaction Remains Responsive

5.4 Loading Behaviour

The prototype shall provide the approved loading behaviour required during the Opening Sequence and relevant transition states.

PR-5.4.1 — Opening Loading

At application launch, the Opening Sequence shall display the approved loading presentation. The loading bar shall progress through 12% → 38% → 67% → 89% → 99% and intentionally stop at 99%.

PR-5.4.2 — 99% Freeze

The loading presentation shall remain frozen at 99% for the defined short period to create the intended impression that something has gone wrong.

PR-5.4.3 — Glitch Transition

After the 99% freeze: the loading bar shall flicker; a digital glitch sound shall play; the background shall distort; the connection status shall change to Connection Failed; and the screen shall cut to complete black.

PR-5.4.4 — Complete Darkness

During the complete-black phase, no UI or loading indicator shall be displayed. The sequence shall remain silent before the ancient forge sound begins.

PR-5.4.5 — Opening Continuation

The opening shall continue from the darkness into the approved Connection Established with The SkillForge SkillForge — Software Requirements Specification | Page 36 SkillForge — Software Requirements Specification | Page 36 transition. This is part of the final Opening Sequence and shall not be treated as a separate scene or sequence.

PR-5.4.6 — Loading Scope

The loading behaviour described here applies to the approved Opening Sequence. No additional loading screen, artificial loading delay, progress percentage or loading system shall be introduced without an approved requirement.

5.4.7 — Loading Flow

Game Launch → Fake Loading → 99% Freeze → Glitch → Connection Failed → Complete Black → Forge Sound → Connection Established with The SkillForge

5.5 Prototype-Level Scalability

The SkillForge prototype shall maintain its intended functionality when operated within its defined prototype scope and environment.

PR-5.5.1 — Prototype Scope

Scalability requirements shall apply only to the prototype level. The system is not required to support commercial-scale deployment or large-scale concurrent users.

PR-5.5.2 — Feature Scalability

The prototype architecture shall allow the approved prototype features to operate together without unnecessary conflicts between the HUB, Challenge System, Forge Vision, Analysis Panels, Reward System, Save System, Audio System and UI Systems.

PR-5.5.3 — Progression Scalability

The progression system shall support the approved prototype progression flow, including player progression, skills, challenges and rewards. No additional large-scale progression or content-generation requirement shall be introduced.

PR-5.5.4 — Data Scalability

The Save System shall support the amount of player and progression data required by the approved prototype. The system shall not be required to support large-scale database infrastructure.

PR-5.5.5 — Content Scalability

The prototype shall support the approved number and scope of prototype content. Future expansion beyond the prototype scope is outside the current scalability requirement.

PR-5.5.6 — Prototype Constraint

Scalability shall be evaluated according to the intended prototype environment and demonstration requirements. No unsupported numerical limits for users, challenges, save size, memory or concurrent sessions shall be introduced.

5.5.7 — Scalability Flow

Prototype Features → Defined Prototype Environment → Systems Operate Together → Stable Prototype Experience

Chapter 6 — Non-Functional Requirements

6.1 Usability

The SkillForge prototype shall provide a clear and understandable interaction experience for its intended users.

NFR-6.1.1 — Interface Understandability

The prototype shall present its approved UI elements and interactions in a manner that allows the player to understand the available actions and current interface state.

NFR-6.1.2 — Interaction Consistency

The prototype shall maintain consistent interaction behaviour across approved UI panels, popups, hints, notifications and gameplay interactions.

NFR-6.1.3 — Navigation

The player shall be able to follow the approved navigation flow between the Guest HUB, Active HUB, Challenge Selection, Challenge Chamber, Reward Popup and return-to-HUB states.

NFR-6.1.4 — Feedback

The prototype shall provide the approved visual or audio feedback for relevant player actions and system events through the defined Notifications, Hints, Analysis Panels and Audio behaviour.

NFR-6.1.5 — Error Prevention

The interface shall prevent or clearly handle invalid interactions according to the approved UI and gameplay rules.

NFR-6.1.6 — Prototype Scope

Usability requirements shall apply to the approved SkillForge prototype and shall not introduce additional accessibility, localization or commercial-product requirements that are not part of the locked project scope.

6.1.7 — Usability Flow

Player Action → Clear Interface Response → Feedback → Next Valid Interaction

6.2 Reliability

The SkillForge prototype shall operate consistently and preserve the approved prototype state and behaviour during normal use.

NFR-6.2.1 — Stable Operation

The prototype shall operate without unnecessary crashes, freezes or interruptions during the approved gameplay flow.

NFR-6.2.2 — State Consistency

The system shall maintain consistent UI, gameplay and progression states while the player moves through the approved prototype flow.

NFR-6.2.3 — Challenge Reliability

The Challenge System shall consistently process the approved Challenge interactions, completion state and Portal transition.

NFR-6.2.4 — Progress Reliability

After successful Challenge completion, the approved progression and reward data shall be processed according to the locked Save System requirements. An incomplete Challenge shall not incorrectly grant or retain completion progress.

NFR-6.2.5 — Recovery from Invalid Interaction

Where an invalid or unsupported interaction occurs, the prototype shall maintain a valid system state rather than producing an unintended progression or transition.

NFR-6.2.6 — Prototype Scope

Reliability requirements shall be evaluated within the intended prototype environment. The prototype shall not be treated as requiring production-scale fault tolerance, server redundancy or disaster-recovery infrastructure.

6.2.7 — Reliability Flow

Player Interaction → System Processes Event → Valid State Maintained → Approved Result / Progression

6.3 Security

The SkillForge prototype shall protect the approved account and persistent player data within the scope of the academic prototype.

NFR-6.3.1 — Account Access

The Login / Sign-Up flow shall restrict account-specific progression access to the appropriate account state. Guest HUB access shall remain separate from the Active HUB account state.

NFR-6.3.2 — Account Data

The prototype shall handle account-related and player progression information only as required by the approved Login, Sign-Up and Save System requirements.

NFR-6.3.3 — Persistent Data Ownership

Persistent player data shall be handled through the approved SaveManager. No other system shall independently perform persistent save operations.

NFR-6.3.4 — Progress Protection

The system shall prevent invalid or unintended progression updates. Reward and progression data shall be applied only through the approved Challenge completion and Save flow.

NFR-6.3.5 — Guest State

Guest users shall not receive unintended access to account-specific progression functionality before account initialization.

NFR-6.3.6 — Prototype Security Scope

Security requirements shall apply to the approved academic prototype. The prototype shall not be required to implement production-level authentication infrastructure, encryption architecture, server-side security or commercial cybersecurity controls unless separately approved.

6.3.7 — Security Flow

Account State → Authorized Interaction → Approved Data Operation → Persistent Data

6.4 Maintainability

The SkillForge prototype shall maintain a structured organization that supports modification and maintenance during prototype development.

NFR-6.4.1 — Modular Organization

The system shall maintain separation between major responsibilities so that changes to one approved system do not unnecessarily affect unrelated systems.

NFR-6.4.2 — Manager Responsibility

The architecture shall follow the locked rule: One Manager = One Responsibility. Managers shall not duplicate ownership of systems or data.

NFR-6.4.3 — Script Responsibility

Scripts shall follow the approved single-responsibility approach. Controllers shall communicate through Managers where required, while unnecessary Controller-to-Controller communication shall be avoided.

NFR-6.4.4 — Data Ownership

The prototype shall maintain clear ownership of data. Persistent data shall be handled by SaveManager, while runtime player data shall belong to PlayerManager.

NFR-6.4.5 — Reusable Components

Reusable objects shall use the approved prefab approach, while unique environment elements may remain as scene objects.

NFR-6.4.6 — Documentation Constraint

The SRS shall document approved maintainability requirements without introducing new technical architecture decisions or implementation details.

6.4.7 — Maintainability Flow

Separated Responsibility → Clear Data Ownership → Modular System → Easier Prototype Maintenance.

6.5 Compatibility

The SkillForge prototype shall operate within the approved target environment defined for the project.

NFR-6.5.1 — Target Platform

The current prototype shall target Windows Desktop.

NFR-6.5.2 — Unity Environment

The prototype shall operate within the approved Unity development/runtime environment used for the project.

NFR-6.5.3 — Scene Compatibility

The prototype shall support the approved two-scene architecture: MainScene and ChallengeScene. Only Challenge entry and return shall require scene transitions; UI-driven progression shall remain within the Main Scene.

NFR-6.5.4 — Input Compatibility

The prototype shall support the approved keyboard and mouse interactions required by the documented gameplay and UI requirements.

NFR-6.5.5 — Feature Compatibility

The approved UI, gameplay, audio, save, progression and challenge systems shall operate together within the defined prototype environment without introducing incompatible system behaviour.

NFR-6.5.6 — Compatibility Scope

Compatibility requirements shall apply to the approved prototype environment. No additional operating systems, mobile platforms, consoles, browsers or hardware configurations shall be treated as current requirements unless separately approved.

6.5.7 — Compatibility Flow

Approved Environment → Unity Prototype → MainScene / ChallengeScene → Approved Systems Operate Together

Chapter 7 — System Architecture

7.1 Technical Architecture Summary

The SkillForge prototype shall use a modular Unity-based architecture designed around clear separation of responsibilities between scenes, managers, controllers, UI components, data classes and reusable objects.

TECH-7.1.1 — Prototype Architecture

The prototype shall use a 2.5D, fixed-camera presentation and shall operate using two Unity Scenes: MainScene and ChallengeScene. The MainScene shall contain the HUB and UI-driven progression, while ChallengeScene shall contain the Challenge environment and puzzle-specific systems.

TECH-7.1.2 — Scene Architecture

Scene transitions shall be used only where required for Challenge entry and return. UI-driven progression shall remain within MainScene. A separate Reward Scene/Reward Chamber shall not be used; the Reward Popup occurs within the Portal transition.

TECH-7.1.3 — Manager Architecture

The architecture shall follow: One Manager = One Responsibility. Managers shall communicate only where required and shall not duplicate ownership of systems or data.

TECH-7.1.4 — Script Architecture

Scripts shall follow One Script = One Responsibility. The architecture separates Manager, Controller, UI, Data, Base Interaction, Animation, Visual Effects and Utility scripts. Controllers shall communicate through Managers, while unnecessary Controller-to-Controller communication shall be avoided. SaveManager shall remain the only persistent save handler.

TECH-7.1.5 — Data Ownership

Persistent data shall be owned by SaveManager, while runtime player data shall belong to PlayerManager. Each data responsibility shall have one clear owner.

TECH-7.1.6 — Reusable Objects

Reusable objects shall use the approved prefab-based approach. Unique environment elements may remain as scene objects.

TECH-7.1.7 — Unity Hierarchy

Each GameObject shall belong to a logical parent. MainScene and ChallengeScene shall contain only the objects required for their respective responsibilities, with runtime-created objects organized under Runtime Objects.

TECH-7.1.8 — Architecture Communication

The architecture shall avoid unnecessary direct dependencies and circular data flow. Controllers communicate through Managers, Managers coordinate with required Managers/UI, and persistent save operations remain centralized in SaveManager.

7.1.9 — Overall Architecture Flow

MainScene / HUB → Managers + UI + Environment → Challenge Entry → ChallengeScene → Challenge Managers + Controllers + Analysis → Portal Transition → Reward Popup → Return to MainScene / HUB

7.2 Scene Architecture

The SkillForge prototype shall use exactly two Unity Scenes: MainScene and ChallengeScene.

TECH-7.2.1 — MainScene

MainScene shall contain the primary HUB environment and the systems required for UI-driven progression. Its approved hierarchy includes Managers, Environment, Main Camera, Canvas, Audio, Runtime Objects and Event System.

TECH-7.2.2 — MainScene Managers

MainScene shall contain the approved managers responsible for Game lifecycle, Environment, UI, Settings, Audio, Save, Player data, Notifications, Hints and Progress.

TECH-7.2.3 — ChallengeScene

ChallengeScene shall contain the Challenge environment and puzzle-specific systems. Its approved hierarchy includes Managers, Challenge Environment, Formation, Analysis UI, Main Camera, Lighting, Audio, Runtime Objects and Event System.

TECH-7.2.4 — Challenge Managers

ChallengeScene shall use ChallengeManager, FormationManager, ForgeVisionManager and PortalManager.

TECH-7.2.5 — Persistent Managers

SaveManager, AudioManager, PlayerManager and SettingsManager shall persist across scene transitions using DontDestroyOnLoad.

TECH-7.2.6 — Non-Persistent Challenge Objects

Formation, Rings, Challenge Environment, Portal Effects and Challenge UI shall be recreated per Challenge.

TECH-7.2.7 — Scene Transition Rule

Only Challenge entry and Challenge return shall require scene transitions. UI-driven progression shall remain within MainScene.

TECH-7.2.8 — Reward Scene Constraint

No separate Reward Scene or Reward Chamber shall exist. The Reward Popup occurs as part of the Portal transition.

7.2.9 — Scene Flow

MainScene / HUB → Challenge Entry → ChallengeScene → Challenge Completion / Portal → Challenge Return → MainScene / HUB

7.3 Manager Architecture

The SkillForge prototype shall use a Manager-based architecture in which each Manager has one clearly defined responsibility.

TECH-7.3.1 — Core Architecture Rule

The architecture shall follow: One Manager = One Responsibility. Managers shall communicate only when required, and responsibilities shall not be duplicated.

TECH-7.3.2 — GameManager

GameManager shall handle global game lifecycle, startup, scene coordination and exit. It shall coordinate scene changes requested by the relevant system.

TECH-7.3.3 — EnvironmentManager

EnvironmentManager shall handle HUB awakening, crystal pulse, hall lighting, floor energy, floating environment, torches and environmental effects.

TECH-7.3.4 — UI and Settings Managers

UIManager shall handle panels, glyphs, popups, ESC behaviour and UI animations. SettingsManager shall handle audio settings, future graphics/gameplay settings, applying settings and save requests.

TECH-7.3.5 — Audio and Save Managers

AudioManager shall handle music, ambient audio, SFX, volume and audio transitions. SaveManager shall handle save, load, auto-save and persistent data.

TECH-7.3.6 — Player and Challenge Managers

PlayerManager shall handle runtime player data, XP, level, inventory and Skill-Abilities. ChallengeManager shall handle Challenge start, restart, exit, completion and reward eligibility.

TECH-7.3.7 — Challenge-System Managers

FormationManager shall handle Ring rotation, alignment, energy flow, core resonance and formation completion. Formation completion shall provide the Challenge result used by ChallengeManager for Challenge completion and reward eligibility. ForgeVisionManager shall handle Forge Vision state, analysis mode, scan updates and Analysis Panels.

PortalManager shall handle Portal activation, interaction, Portal Warp initiation and scene transition requests;

GameManager shall coordinate the resulting scene transition.

TECH-7.3.8 — Feedback and Progress Managers

NotificationManager shall handle the Notification queue and message handling. HintManager shall handle tutorial hints, first-time hints and hint progression. ProgressManager shall handle progression-related UI and display state, including Level, XP, Play Time, Reward Claimed and Challenge Indicators. The underlying runtime Level and XP data remain owned by PlayerManager.

7.3.9 — Manager Responsibility Flow

Player / UI Event → Relevant Manager → Required System Operation → Approved Result / State Update

7.4 Script Architecture

The SkillForge prototype shall use a responsibility-based script architecture in which each script has one clear purpose.

TECH-7.4.1 — Core Script Rule

The architecture shall follow: One Script = One Responsibility. Script names shall clearly describe their purpose.

TECH-7.4.2 — Manager Scripts

The approved Manager Scripts shall include: GameManager.cs, EnvironmentManager.cs, UIManager.cs, SettingsManager.cs, AudioManager.cs, SaveManager.cs, PlayerManager.cs, ChallengeManager.cs, FormationManager.cs, ForgeVisionManager.cs, PortalManager.cs, NotificationManager.cs, HintManager.cs and ProgressManager.cs.

TECH-7.4.3 — Controller Scripts

HUB Controllers: CrystalController.cs, ForgeController.cs, PortalController.cs and KnowledgeBookController.cs.

Challenge Controllers: RingController.cs, FormationController.cs, EnergyFlowController.cs and CoreController.cs.

TECH-7.4.4 — UI Scripts

The approved UI Scripts shall include: LoginPanel.cs, ProfilePanel.cs, SkillTreePanel.cs, ForgeCorePanel.cs, ChallengeSelectionPanel.cs, LevelSelectionPanel.cs, SettingsPanel.cs, RewardPopup.cs, NotificationUI.cs, HintUI.cs and ExitPopup.cs.

TECH-7.4.5 — Data Classes

The approved Data Classes shall include: PlayerData.cs, SettingsData.cs, ChallengeData.cs, RewardData.cs, SkillData.cs and SaveData.cs.

TECH-7.4.6 — Base Interaction

InteractableObject.cs shall serve as the base class for Portal, Crystal, Forge Structure, Ancient Knowledge Book SkillForge — Software Requirements Specification | Page 43 SkillForge — Software Requirements Specification | Page 43 and future interactable world objects.

TECH-7.4.7 — Animation and Visual Effects

Approved Animation Scripts: PanelAnimator.cs, PortalAnimator.cs, CrystalAnimator.cs and EnvironmentAnimator.cs. There shall be no separate AnimationManager. Approved Visual Effects: ForgeVisionEffect.cs, EnergyPulseEffect.cs and PortalEffect.cs.

TECH-7.4.8 — Utility and Communication Rules

Approved Utility Scripts: SceneLoader.cs, TimerUtility.cs and AudioUtility.cs. Controllers shall communicate through

Managers. Managers shall coordinate with other Managers and UI. Controller-to-Controller communication should be avoided. SaveManager shall remain the only persistent save handler.

TECH-7.4.9 — Script Expansion Rule

Core scripts are fixed in the architecture. Helper or feature-specific scripts may be added during implementation without breaking the defined architecture or responsibilities.

7.4.10 — Script Communication Flow

Controller / UI → Relevant Manager → Required System / Data → Approved Result

7.5 Prefab Architecture

The SkillForge prototype shall use prefabs for reusable objects to maintain consistency, reusability and easier future expansion.

TECH-7.5.1 — Prefab Philosophy

Anything reusable should be a Prefab. Reusable objects shall be converted into prefabs rather than unnecessarily duplicated as independent scene objects.

TECH-7.5.2 — Environment Prefabs

Approved Environment Prefabs: Forge.prefab, Crystal.prefab, Portal.prefab and KnowledgeBook.prefab. The modular Hall shall use Floor, Wall, Pillar, Ceiling, DecorativeStone, Torch and FloatingIsland prefabs.

TECH-7.5.3 — Gameplay Prefabs

Approved Gameplay Prefabs: Formation.prefab, OuterRing.prefab, MiddleRing.prefab, InnerRing.prefab, EnergyBeam.prefab and EnergyNode.prefab.

TECH-7.5.4 — UI Prefabs

Reusable UI Prefabs: Glyph.prefab, AnalysisPanel.prefab, ProfilePanel.prefab, SkillTreePanel.prefab, ForgeCorePanel.prefab, ChallengeSelectionPanel.prefab, LevelSelectionPanel.prefab, SettingsPanel.prefab, LoginPopup.prefab, RewardPopup.prefab, ExitPopup.prefab, Notification.prefab and Hint.prefab.

TECH-7.5.5 — Effect Prefabs

Approved Effect Prefabs: ForgeVisionScan.prefab, EnergyPulse.prefab, CrystalGlow.prefab, PortalWarp.prefab, HologramEffect.prefab and EnergySpark.prefab.

TECH-7.5.6 — Camera and Lighting Prefabs

MainCameraRig.prefab shall contain the Main Camera and Audio Listener. Approved Lighting Prefabs are PointLight.prefab, CrystalLight.prefab and TorchLight.prefab.

TECH-7.5.7 — Runtime Objects

Notifications, Reward Popup, Login Popup, Exit Popup, Hint UI and temporary energy effects shall be instantiated at runtime.

TECH-7.5.8 — Scene Objects

Hall, Forge, Crystal, Portal and major environment structures shall remain permanently placed in their respective scenes.

TECH-7.5.9 — Nested Prefabs

Nested prefabs are allowed to maximize reuse, such as CrystalGlow and PortalWarp inside larger prefabs.

TECH-7.5.10 — Prefab Rules

Reusable objects become prefabs; unique environment elements remain scene objects; runtime UI is instantiated; VFX use reusable prefabs; and future systems should reuse existing prefabs whenever possible.

7.5.11 — Prefab Flow

Reusable Object → Prefab → Scene / Runtime Instance → Consistent Behaviour

7.6 Folder Structure

The SkillForge project shall use a structured folder hierarchy in which every asset has one permanent and logical location.

TECH-7.6.1 — Folder Philosophy

Every asset has one permanent home. Duplicate assets and random folders shall be avoided.

TECH-7.6.2 — Root Structure

The approved root folders shall include: Assets, Scenes, Scripts, Prefabs, Art, Environment, Animations, Audio, VFX, UI, Config, Documentation and ThirdParty.

TECH-7.6.3 — Scene Structure

The project shall contain only MainScene.unity and ChallengeScene.unity.

TECH-7.6.4 — Script Structure

Scripts shall be organized into Managers, Controllers, UI, Data, Utilities and Effects.

TECH-7.6.5 — Prefab Structure

Prefabs shall be organized into Environment, Gameplay, UI, Effects and Lighting.

TECH-7.6.6 — Art and Environment Structure

Art shall contain Models, Materials, Textures, Sprites and Icons. Environment shall contain the modular Hall assets including Floor, Wall, Pillar, Ceiling, Decorative Structures, Floating Islands and Torches.

TECH-7.6.7 — Animation, Audio and UI Structure

Animations shall contain UI, Portal, Crystal, Forge and Environment animation assets. Audio shall contain Music, Ambient, SFX and UI audio. UI shall contain Fonts, Icons, Sprites and Panels.

TECH-7.6.8 — Configuration Structure

The approved Config folder shall contain GameConfig, AudioConfig, ChallengeConfig, UIConfig and SaveConfig (future-ready).

TECH-7.6.9 — Documentation and ThirdParty

Documentation shall contain SRS, GDD, Flowcharts, DFD, ERD, UML, Prototype Notes, Presentation, Assets Reference and Version History. ThirdParty shall contain external Models, Audio, Fonts, Materials and Textures separately from original project assets.

TECH-7.6.10 — Naming Rules

Folders shall use PascalCase, while scripts, prefabs, materials and textures shall use descriptive names.

7.6.11 — Folder Organization Flow

Asset → Correct Category → Permanent Folder → No Duplicate / Random Location

7.7 Data Flow

The SkillForge prototype shall use a controlled data flow in which runtime data, persistent data and system communication follow clearly defined ownership.

TECH-7.7.1 — Data Ownership

Each data responsibility shall have one clear owner. PlayerManager handles runtime player data; SaveManager handles persistent data; ChallengeManager handles Challenge state; ProgressManager handles progression display/state;

SettingsManager handles settings state and save requests. ProgressManager does not become a second owner of Player runtime data.

TECH-7.7.2 — Player Data Flow

Runtime player information shall be managed through PlayerManager, including XP, Level, Inventory and

Skill-Abilities.

TECH-7.7.3 — Persistent Data Flow

Persistent data shall be handled through SaveManager, including Save, Load, Auto-save and persistent data.

TECH-7.7.4 — Challenge Data Flow

Challenge interaction shall be coordinated through ChallengeManager, while puzzle-specific operations shall be handled by the appropriate Challenge Managers. ChallengeManager handles Challenge start, restart, exit, completion and reward eligibility.

TECH-7.7.5 — Formation Data Flow

Formation-related gameplay data shall be handled by FormationManager, including Ring rotation, alignment, energy flow, Core Resonance and formation completion.

TECH-7.7.6 — Forge Vision Data Flow

Forge Vision analysis shall be managed through ForgeVisionManager, including Forge Vision state, analysis mode, scan updates and Analysis Panels.

TECH-7.7.7 — Progression Data Flow

Progress-related information shall be presented through ProgressManager, including Progress UI, Level, XP, Play Time, Reward Claimed and Challenge Indicators. The underlying runtime Level and XP data remain owned by PlayerManager.

TECH-7.7.8 — Settings Data Flow

SettingsManager shall handle settings, apply settings and save requests, while SaveManager remains responsible for persistent storage.

TECH-7.7.9 — Communication Rule

Controllers shall communicate through Managers. Managers shall coordinate with other required Managers and UI.

Controller-to-Controller communication should be avoided. SaveManager shall remain the only persistent save handler.

7.7.10 — Overall Data Flow

Player / UI / Gameplay Event → Relevant Manager → Runtime Data / System State → SaveManager when persistence is required → Persistent Data

7.8 Unity Object Hierarchy

The SkillForge Unity project shall use a logical GameObject hierarchy in which every GameObject belongs to an appropriate parent.

TECH-7.8.1 — Hierarchy Philosophy

Every GameObject shall belong to a logical parent to keep the project clean, scalable and easy to debug.

TECH-7.8.2 — MainScene Hierarchy

MainScene shall contain Managers, Environment, Main Camera, Canvas, Audio, Runtime Objects and Event System.

TECH-7.8.3 — MainScene Managers

MainScene Managers shall include GameManager, EnvironmentManager, UIManager, SettingsManager,

AudioManager, SaveManager, PlayerManager, NotificationManager, HintManager and ProgressManager.

TECH-7.8.4 — ChallengeScene Hierarchy

ChallengeScene shall contain Managers, Challenge Environment, Formation, Analysis UI, Main Camera, Lighting, Audio, Runtime Objects and Event System.

TECH-7.8.5 — Challenge Managers

ChallengeScene Managers shall include ChallengeManager, FormationManager, ForgeVisionManager and PortalManager.

TECH-7.8.6 — Persistent Managers

SaveManager, AudioManager, PlayerManager and SettingsManager shall use DontDestroyOnLoad.

TECH-7.8.7 — Non-Persistent Objects

Formation, Rings, Challenge Environment, Portal Effects and Challenge UI shall be recreated per Challenge.

TECH-7.8.8 — Runtime Object Rule

Runtime-created objects shall be parented under Runtime Objects. Each scene shall contain only the objects required for that scene.

TECH-7.8.9 — Naming Rule

GameObjects shall use meaningful names. Default Unity names such as Cube or GameObject shall be avoided.

7.8.10 — Hierarchy Flow

Scene → Logical Parent → System / Environment / UI Object → Child Objects

Chapter 8 — Data Management

8.1 Player Data

The SkillForge prototype shall maintain the player information required for player identity, progression and runtime gameplay state.

DATA-8.1.1 — Player Identity

The system shall maintain the player's Username and Player ID as the approved player identity information.

DATA-8.1.2 — Player Progress State

The system shall maintain the player's current Level and Total XP as part of the player progression state.

DATA-8.1.3 — Runtime Player Data

During gameplay, PlayerManager shall handle runtime player data including XP, Level, Inventory and Skill-Abilities.

DATA-8.1.4 — Player State and Account State

The player may exist in either of the approved progression states: Guest User or Authenticated User. These are not separate player roles; they are two states of the same primary player.

DATA-8.1.5 — New Account State

For a new account, the approved initial state is Level 1, XP 0, empty inventory, no completed challenges and Forge Vision locked.

DATA-8.1.6 — Data Ownership

Runtime player data shall be owned by PlayerManager. Persistent storage shall remain the responsibility of SaveManager.

DATA-8.1.7 — Data Scope

Only player information required by the approved prototype shall be maintained. No additional personal profile fields, social information, administrative data or multiplayer-player data shall be introduced.

8.1.8 — Player Data Flow

Player / Account State → Player Identity → PlayerManager → Runtime Player Data → SaveManager when persistent storage is required

8.2 Save Data

The SkillForge prototype shall maintain the approved player and progression information through the defined Save System.

DATA-8.2.1 — Save Ownership

SaveManager shall be responsible for Save, Load, Auto-save and Persistent Data and shall remain the only persistent save handler.

DATA-8.2.2 — Initial Save State

A new account shall begin with Level 1, XP 0, empty inventory, no completed challenges and Forge Vision locked.

DATA-8.2.3 — Challenge Save Rule

The system shall not save Challenge progress before successful completion. After successful Challenge completion, the approved completion and progression state shall be saved.

DATA-8.2.4 — Completion Save

Successful Challenge completion shall update the appropriate persistent progression state. The completion state shall not be incorrectly lost when the player exits after completing the Challenge.

DATA-8.2.5 — Auto-Save

The approved Save System shall support auto-save through SaveManager. No additional trigger interval or storage mechanism shall be assumed where it is not defined by the locked source.

DATA-8.2.6 — Restart Behaviour

Restarting a Challenge shall reset the current Challenge state rather than incorrectly treating an incomplete attempt as completed progress.

DATA-8.2.7 — Reward Persistence

After successful Challenge completion, the approved reward and progression state shall be persisted according to the Save System.

DATA-8.2.8 — Persistent Data Scope

Only player and progression information required by the approved prototype shall be persisted. No additional cloud-sync, multiplayer-save or backup infrastructure shall be assumed.

8.2.9 — Save Data Flow

Challenge / Player Event → Completion / Valid Save Condition → SaveManager → Persistent Player Data

8.3 Progress Data

The SkillForge prototype shall maintain the progression information required to represent the player's current game progress.

DATA-8.3.1 — Level

The system shall maintain the player's current Level as part of progression data.

DATA-8.3.2 — Experience

The system shall maintain the player's XP and update progression according to the approved Challenge and reward flow.

DATA-8.3.3 — Challenge Progress

The system shall maintain the approved Challenge completion state. An incomplete Challenge shall not incorrectly be treated as completed progression.

DATA-8.3.4 — Reward Progress

The system shall maintain whether the approved Challenge reward has been claimed as part of the progression state.

DATA-8.3.5 — Progress UI Data

ProgressManager shall handle progression-related UI and display state, including Level, XP, Play Time, Reward Claimed and Challenge Indicators. The underlying runtime Level and XP data remain owned by PlayerManager.

DATA-8.3.6 — Skill Progression

The approved player progression shall support the player's Skill-Abilities as part of the runtime player state.

DATA-8.3.7 — Progress Persistence

Progression that is required to persist shall be handled through SaveManager after the approved save condition.

DATA-8.3.8 — New Player Progress

The approved initial progression state for a new account is: Level 1, XP 0, No Completed Challenges and Forge Vision Locked.

8.3.9 — Progress Data Flow

Challenge Completion / Reward → PlayerManager (Runtime Player Data Update) → ProgressManager (Progress Display Update) → SaveManager (Persistent Save) → Persistent Progress

8.4 Reward Data

The SkillForge prototype shall maintain the reward information required for successful Challenge completion and approved progression.

DATA-8.4.1 — Reward Eligibility

A player shall become eligible for the approved Challenge reward only after the Challenge has been successfully completed.

DATA-8.4.2 — Reward State

The system shall maintain whether the approved Challenge reward has been claimed. This state shall be reflected in the player's progression data.

DATA-8.4.3 — Reward Processing

ChallengeManager shall handle Challenge completion and reward eligibility.

DATA-8.4.4 — Reward Persistence

After successful Challenge completion, the approved reward and progression state shall be persisted through SaveManager according to the Save System.

DATA-8.4.5 — Reward and Progression

Reward processing shall update only the progression information approved for the prototype. The reward shall not be granted for an incomplete Challenge.

DATA-8.4.6 — Reward Display

The approved Reward Popup shall present the reward/result to the player as part of the Challenge completion flow.

The Reward Popup is part of the Portal transition and does not require a separate Reward Scene.

DATA-8.4.7 — Reward Data Scope

Only reward information required by the approved prototype shall be maintained. No additional reward types, currencies, inventories or economy systems shall be introduced unless already approved.

8.4.8 — Reward Data Flow

Challenge Completed → Reward Eligibility → Reward Popup → Reward Claimed / Progression Update → SaveManager

8.5 Settings Data

The SkillForge prototype shall maintain the approved settings information required for configuration and persistence of the player's configured state.

DATA-8.5.1 — Settings Ownership

SettingsManager shall be responsible for Audio Settings, future Graphics/GamePlay Settings, applying settings and Save Requests.

DATA-8.5.2 — Audio Settings

The system shall maintain the approved audio settings required by the prototype. These settings shall be handled through SettingsManager and applied through the appropriate audio system.

DATA-8.5.3 — Future Settings

The architecture shall remain capable of supporting future graphics/gameplay settings through SettingsManager.

Such future settings shall not be treated as currently required prototype features unless separately approved.

DATA-8.5.4 — Settings Application

When the player changes an approved setting, SettingsManager shall apply the setting according to the defined system behaviour.

DATA-8.5.5 — Settings Persistence

SettingsManager shall issue save requests when settings require persistence, while SaveManager remains responsible for the actual persistent save operation.

DATA-8.5.6 — Settings Scope

Only settings defined by the approved prototype shall be maintained. No additional graphics, gameplay, accessibility or platform-specific settings shall be added without an approved requirement.

DATA-8.5.7 — Settings State

The system shall maintain the current values of approved settings required to restore the player's configured state when persistence is applicable.

8.5.8 — Settings Data Flow

Player Changes Setting → SettingsManager → Apply Setting → Save Request (if required) → SaveManager

Chapter 9 — Testing Requirements

9.1 Feature Testing

Feature testing shall verify that the approved SkillForge prototype features operate according to their documented functional requirements.

TEST-9.1.1 — Opening Experience

Test the approved Opening Sequence. Expected Result: The sequence shall proceed through its approved flow and transition the player into the Hall/HUB experience.

TEST-9.1.2 — HUB & Account Flow

Test entry to the Guest HUB and the approved Login / Sign-Up flow. Expected Result: The player shall be able to access the account-related flow and proceed to the authenticated progression state.

TEST-9.1.3 — Data Feed & Glyph Conversion

Test the Data Feed and Glyph Conversion after the required account flow. Expected Result: The approved progression data shall become available and Glyph Conversion shall operate according to the defined prototype flow.

TEST-9.1.4 — Forge Vision & Challenge Access

Test the required Forge Vision progression and Challenge access condition. Expected Result: Challenge access shall become available only after the required progression condition is satisfied.

TEST-9.1.5 — Challenge Puzzle Flow

Test the approved Challenge sequence including Ring System, Alignment Analysis, Energy Flow and Core

Resonance. Expected Result: The required stages shall operate and allow successful Challenge completion.

TEST-9.1.6 — Portal Activation

Test Portal interaction after successful Challenge completion. Expected Result: The Portal shall activate according to the approved completion flow.

TEST-9.1.7 — Reward Presentation

Test the reward flow after successful Challenge completion. Expected Result: The approved Reward Popup/reward presentation shall occur and Challenge progression shall be completed.

TEST-9.1.8 — Save & Progress Behaviour

Test the resulting progression after Challenge completion. Expected Result: Approved completion/progression data shall be saved, while incomplete Challenge state shall not incorrectly persist as completed progress.

TEST-9.1.9 — Supporting Systems

Test the approved Notifications, Hints, Settings and required interface interactions during the feature flow.

Expected

Result: Each supporting feature shall respond according to its approved behaviour without disrupting the primary gameplay flow.

9.1.10 — Feature Testing Result Rule

Each feature test shall be recorded as PASS when the expected behaviour is achieved or FAIL when the expected behaviour is not achieved. Results shall only be marked after the actual Unity prototype has been tested.

9.1.11 — Feature Testing Flow

Approved Feature → Execute Feature → Observe Behaviour → Compare with Requirement → PASS / FAIL

9.2 UI Testing

UI testing shall verify that approved SkillForge interface elements, panels, interactions and UI flow operate according to their documented requirements.

TEST-9.2.1 — HUB Interface

Test the approved Guest HUB / Active HUB interface and its available interactions. Expected Result: Required HUB elements shall be visible and usable according to the approved interface behaviour.

TEST-9.2.2 — Challenge Interface

Enter the Challenge Chamber and verify its approved interface. Expected Result: Challenge-related controls, visual elements and required interaction areas shall be available without interfering with gameplay.

TEST-9.2.3 — Analysis Panels

Trigger the approved Analysis flow during the Challenge. Expected Result: Analysis Panels shall appear and present the required analysis information according to the approved behaviour.

TEST-9.2.4 — Reward Popup

Complete the Challenge and verify the Reward Popup. Expected Result: The Reward Popup shall appear at the approved point in the completion flow and provide the required reward/result information.

TEST-9.2.5 — Notification System

Trigger approved system/gameplay events that require notifications. Expected Result: Notifications shall appear with the approved behaviour and shall not unnecessarily obstruct the primary interaction flow.

TEST-9.2.6 — Hint System

Use the approved Hint System during the relevant Challenge interaction. Expected Result: The Hint System shall provide the approved hint behaviour without incorrectly completing or bypassing the Challenge.

TEST-9.2.7 — Settings Panel

Open the Settings Panel and modify an approved setting. Expected Result: The setting shall respond according to the approved Settings behaviour and the panel shall remain usable.

TEST-9.2.8 — Panel Behaviour

Open, close and interact with approved panels/popups throughout the prototype. Expected Result: Panels shall follow their approved visibility, interaction and navigation behaviour.

TEST-9.2.9 — UI Flow

Follow the complete approved UI-driven navigation flow from HUB through Challenge/Reward and back to HUB. Expected Result: The player shall be able to follow the approved UI flow without unintended panel states or navigation breaks.

9.2.10 — UI Testing Result Rule

Each UI test shall be recorded as PASS or FAIL only after testing the actual Unity prototype.

9.2.11 — UI Testing Flow

UI Element / Panel → User Interaction → Visual/UI Response → Compare with Requirement → PASS / FAIL

9.3 Gameplay Testing

Gameplay testing shall verify that the approved gameplay mechanics and Challenge interactions operate according to their documented requirements.

TEST-9.3.1 — Challenge Entry

Enter the Challenge through the approved Challenge access flow. Expected Result: The Challenge shall start in the approved initial Challenge state.

TEST-9.3.2 — Ring System

Interact with the Challenge Rings according to the approved puzzle flow. Expected Result: Ring rotation and alignment behaviour shall operate as specified.

TEST-9.3.3 — Alignment Analysis

Perform the approved Alignment Analysis. Expected Result: The system shall provide the required analysis behaviour and shall correctly reflect the current alignment state.

TEST-9.3.4 — Energy Flow

Complete the required Ring alignment and observe Energy Flow. Expected Result: Energy shall flow according to the approved Challenge sequence.

TEST-9.3.5 — Core Resonance

Complete the required Energy Flow and trigger Core Resonance. Expected Result: Core Resonance shall occur according to the approved Challenge completion sequence.

TEST-9.3.6 — Forge Vision

Use Forge Vision during the approved Challenge interaction. Expected Result: Forge Vision state, analysis mode and scan behaviour shall operate according to the approved requirements.

TEST-9.3.7 — Challenge Completion

Complete all required Challenge stages successfully. Expected Result: The Challenge shall be recognized as successfully completed and proceed to the approved completion/Portal flow.

TEST-9.3.8 — Challenge Restart

Restart an active Challenge. Expected Result: The Challenge shall restart from the approved Starting Chamber state rather than treating the previous incomplete attempt as completed progress.

TEST-9.3.9 — Challenge Exit

Exit the Challenge through the approved exit behaviour. Expected Result: The player shall leave the active Challenge without incorrectly recording an incomplete attempt as completed progression.

TEST-9.3.10 — Gameplay Result Rule

Each gameplay test shall be recorded as PASS or FAIL only after testing the actual prototype.

9.3.11 — Gameplay Testing Flow

Challenge Start → Gameplay Interaction → Puzzle State Change → Completion / Exit / Restart → Compare with Requirement → PASS / FAIL.

9.4 Save Testing

Save testing shall verify that the approved Save System correctly handles saving, loading, auto-saving and persistence of approved player and progression data.

TEST-9.4.1 — New Player Save State

Create/access a new player state and inspect the initial progression state. Expected Result: Level 1, XP 0, empty inventory, no completed Challenges and Forge Vision locked.

TEST-9.4.2 — Save Operation

Trigger an approved condition requiring persistent player/progression data. Expected Result: SaveManager shall store the approved persistent state.

TEST-9.4.3 — Load Operation

Save valid progression data, leave the relevant flow and load the player state again. Expected Result: The approved saved progression state shall be restored correctly.

TEST-9.4.4 — Auto-Save

Perform an approved action that requires auto-save. Expected Result: The relevant persistent state shall be handled through SaveManager's auto-save functionality. No specific auto-save interval is assumed because the locked source does not define one.

TEST-9.4.5 — Challenge Completion Persistence

Successfully complete a Challenge and verify the player's saved progression. Expected Result: The approved completion/progression state shall persist after successful completion.

TEST-9.4.6 — Incomplete Challenge Protection

Exit or restart a Challenge before successful completion and inspect persistent progression. Expected Result: The incomplete attempt shall not be incorrectly recorded as completed Challenge progression.

TEST-9.4.7 — Restart Save Behaviour

Restart an active Challenge and inspect progression. Expected Result: Restart shall reset the current Challenge state without incorrectly creating completed persistent progression.

TEST-9.4.8 — Reward Persistence

Complete the approved Challenge reward flow and verify saved progression. Expected Result: The approved reward/progression state shall remain available after the save operation.

TEST-9.4.9 — Save Ownership

Verify persistent save operations during the prototype flow. Expected Result: Persistent save handling shall remain centralized through SaveManager; another Manager shall not independently perform persistent saving. **9.4.10 — Save Testing Result Rule**

Each save test shall be marked PASS or FAIL only after the actual prototype has been tested.

9.4.11 — Save Testing Flow

Gameplay / Progress Event → Valid Save Condition → SaveManager → Persistent State → Load / Verify → PASS / FAIL

9.5 Audio Testing

Audio testing shall verify that the approved Audio System operates according to its documented requirements. **TEST-9.5.1 — Background Music**

Launch the prototype and navigate through the approved game flow. Expected Result: Background music shall play according to the approved audio behaviour and transition appropriately where specified.

TEST-9.5.2 — Ambient Audio

Enter the approved HUB/environment and Challenge environment. Expected Result: Appropriate ambient audio shall play according to the approved environment/audio behaviour.

TEST-9.5.3 — Sound Effects

Trigger approved gameplay and UI interactions that use SFX. Expected Result: Corresponding sound effects shall play at the appropriate interaction points.

TEST-9.5.4 — UI Audio

Interact with approved UI elements that have associated audio feedback. Expected Result: Required UI audio feedback shall occur without incorrect or repeated playback.

TEST-9.5.5 — Volume Settings

Modify the approved audio settings through the Settings Panel. Expected Result: The corresponding audio behaviour shall respond to the changed setting.

TEST-9.5.6 — Audio Transitions

Move through approved scene/gameplay transitions where audio changes are required. Expected Result: Audio transitions shall occur without unintended abrupt behaviour or conflicting audio playback.

TEST-9.5.7 — AudioManager Ownership

Verify audio operations throughout the prototype. Expected Result: Music, ambient audio, SFX, volume and audio transitions shall be coordinated through AudioManager according to the approved architecture.

9.5.8 — Audio Testing Result Rule

Each audio test shall be marked PASS or FAIL only after testing the actual Unity prototype.

9.5.9 — Audio Testing Flow

Audio Event / Setting → AudioManager / Approved Audio System → Expected Audio Response → PASS / FAIL

9.6 Transition Testing

Transition testing shall verify that approved scene, gameplay and UI transitions operate according to the documented prototype flow.

TEST-9.6.1 — Opening Sequence Transition

Execute the complete Opening Sequence. Expected Result: The sequence shall transition correctly into the approved Hall/HUB experience without an unintended break in flow.

TEST-9.6.2 — HUB → Challenge Transition

Access an available Challenge from the approved HUB flow. Expected Result: The prototype shall transition from MainScene to ChallengeScene correctly.

TEST-9.6.3 — Challenge Initial State

Enter ChallengeScene. Expected Result: The Challenge shall begin in its approved initial state and required Challenge objects and systems shall be available.

TEST-9.6.4 — Challenge → Portal Transition

Successfully complete the approved Challenge sequence and activate the Portal. Expected Result: The Portal transition shall execute according to the approved Challenge completion flow.

TEST-9.6.5 — Reward Transition

Complete the Portal transition and observe the reward flow. Expected Result: The approved Reward Popup shall appear as part of the Portal transition. No separate Reward Scene shall be expected.

TEST-9.6.6 — Challenge → MainScene Return

Complete the approved Challenge return flow. Expected Result: The prototype shall return from ChallengeScene to MainScene/HUB correctly.

TEST-9.6.7 — Persistent Manager Continuity

Verify persistent systems across the MainScene ↔ ChallengeScene transition. Expected Result: SaveManager, AudioManager, PlayerManager and SettingsManager shall remain available across scene transitions according to the approved architecture.

TEST-9.6.8 — UI Flow Continuity

Follow the approved UI-driven flow before and after Challenge transitions. Expected Result: UI state and navigation shall continue according to the approved flow without unintended transition breaks.

9.6.9 — Transition Result Rule

Each transition test shall be recorded as PASS or FAIL only after testing the actual Unity prototype.

9.6.10 — Transition Testing Flow

MainScene / HUB → Challenge Entry → ChallengeScene → Challenge Completion → Portal Transition → Reward Popup → MainScene / HUB

Chapter 10 — Acceptance Criteria

10.1 Opening Works

The Opening Sequence shall be considered accepted when the complete approved introductory experience executes successfully from application launch through the Hall reveal and the “Press ENTER to Begin” prompt.

AC-10.1.1 — Opening Sequence Launch

When the application is launched, the approved Opening Sequence shall begin automatically. The initial futuristic interface, connection message and loading presentation shall appear as defined.

AC-10.1.2 — Fake Loading Sequence

The simulated loading sequence shall progress through the approved stages: 12% → 38% → 67% → 89% → 99%, and shall intentionally stop at 99%.

AC-10.1.3 — Glitch & Connection Failure

After the 99% freeze, the approved glitch transition shall occur, including loading flicker, digital glitch sound and visual distortion. The connection status shall change to “Connection Failed”, followed by the complete black-screen phase.

AC-10.1.4 — Connection Established

The Opening Sequence shall continue from the black-screen/forge-sound phase into the approved “Connection Established with The SkillForge” transition. This shall remain part of the single approved Opening Sequence, not a separate scene or sequence.

AC-10.1.5 — Crystal & Narrative Reveal

The blue crystal reveal, approved narrative presentation and Crystal Awakening sequence shall execute according to the locked Opening Sequence requirements.

AC-10.1.6 — Logo & Hall Reveal

The SkillForge logo shall appear after the awakening sequence, followed by the approved Hall reveal containing the required environment elements.

AC-10.1.7 — Opening Completion

After the Hall has been revealed, the system shall display “Press ENTER to Begin” and the Opening Sequence shall be considered successfully completed.

AC-10.1.8 — Overall Acceptance Rule

Opening Works = PASS only when the complete approved sequence executes without a blocking defect from:

Launch → Fake Loading → 99% Freeze → Glitch → Connection Failed → Black Screen → Forge Sound → Connection Established → Crystal Reveal → Narrative → Crystal Awakening → Logo → Hall Reveal → ENTER Prompt.

10.2 HUB Works

The HUB shall be considered accepted when the approved Guest HUB, account transition, Active HUB, panel interaction and progression flows execute correctly within the MainScene.

AC-10.2.1 — HUB Entry

After the Opening Sequence, pressing ENTER shall initiate the approved HUB Awakening flow. The Forge shall awaken, energy shall flow to the Blue Crystal, an Energy Pulse shall be released and the approved HUB elements shall activate into the awakened state.

AC-10.2.2 — Guest HUB State

The initial Guest HUB shall display the approved HUB panels. Gameplay data shall be absent and the Profile Panel shall be the available account interaction. The player shall proceed through Login/Sign-Up before using the Challenge Portal.

AC-10.2.3 — Account Transition to Active HUB

After Login/Sign-Up, default player data shall be generated, required panel data shall become available and the approved panels shall convert to their Glyph Form.

AC-10.2.4 — HUB Core Elements

The approved HUB elements shall be available in their defined positions and behaviour, including Player Profile, Settings, Challenge Portal, Forge Core, Skill Tree, Knowledge Module, Inventory, Achievements and Stats/Progression.

AC-10.2.5 — HUB Panel Interaction

Interactive HUB Glyphs shall expand into their associated panels at the same position. Individual Back shall return only that panel to Glyph Form; ESC shall close currently open panels to Glyph Form. Multiple information panels may remain open simultaneously and shall retain independent interaction.

AC-10.2.6 — HUB Visual Behaviour

The HUB shall use the approved fixed-camera presentation without normal camera movement, zoom or rotation. Interactive objects shall provide their approved hover behaviour and the environment shall remain appropriately visible.

AC-10.2.7 — HUB Progression Access

The Active HUB shall provide the approved progression flow through Forge Core, Forge Vision, Skill Tree, Challenge Portal and Challenge Selection according to the required progression state.

AC-10.2.8 — HUB Return

After successful Challenge and Reward completion, the player shall return to the HUB through the approved Portal Warp and 1-second Black Fade. Relevant panels shall update and the Notification Queue shall receive the approved event.

AC-10.2.9 — Overall Acceptance Rule

HUB Works = PASS only when the approved flow executes without a blocking defect: ENTER → HUB Awakening → Guest HUB → Login/Sign-Up → Data Generated → Glyph Conversion → Active HUB → Panel Interaction → Progression Access → Challenge/Reward Return → Updated HUB.

10.3 Login Works

The account flow shall be considered accepted when the approved Guest User to Authenticated User transition operates correctly and provides the required player data for Active HUB progression.

AC-10.3.1 — Guest Account State

The player shall initially access the approved Guest User state. Gameplay data shall be absent and the approved Profile/Login interaction shall be available.

AC-10.3.2 — Login Access

The player shall be able to access the approved Login flow from the Guest state and enter the required authentication information.

AC-10.3.3 — Sign-Up Access

A new player shall be able to access and complete the approved Sign-Up flow and proceed to the authenticated progression state.

AC-10.3.4 — Authenticated State

After successful Login or Sign-Up, the player shall transition to the Authenticated User state. Approved player data shall become available and Active HUB functionality shall be accessible.

AC-10.3.5 — Default Player Data

A newly created account shall begin with Level 1, XP 0, Empty Inventory, No Completed Challenges and Forge Vision Locked.

AC-10.3.6 — Data Feed After Authentication

After authentication, the approved Data Feed process shall execute. Required player data shall become available to HUB panels and the approved Glyph Conversion flow shall proceed.

AC-10.3.7 — Active HUB Transition

After Login or Sign-Up, the player shall proceed to the Active HUB state. Required panel data shall be available and approved panel Glyph Forms shall activate.

AC-10.3.8 — Login Acceptance Rule

Login Works = PASS only when the approved flow completes without a blocking defect: Guest User → Login/Sign-Up → Authentication → Default/Player Data → Data Feed → Glyph Conversion → Active HUB.

10.4 Challenge Playable

The Challenge shall be considered accepted when the approved Challenge progression and puzzle sequence can be completed successfully from Challenge access through Portal activation.

AC-10.4.1 — Challenge Access

After the required progression, the player shall be able to access the approved Challenge Portal and Challenge Selection. The required Challenge shall become available and ChallengeScene shall open correctly.

AC-10.4.2 — Challenge Initial State

When the Challenge starts, the approved initial Challenge state shall load. The Starting Chamber and required Challenge systems and objects shall be available.

AC-10.4.3 — Ring System

The player shall be able to perform the approved Ring System interaction. Ring rotation and alignment behaviour shall respond correctly.

AC-10.4.4 — Alignment Analysis

Required Ring alignment shall allow the approved Alignment Analysis to execute and reflect the current alignment state.

AC-10.4.5 — Energy Flow

After the required alignment, Energy Flow shall execute according to the approved Challenge sequence.

AC-10.4.6 — Core Resonance

After the required Energy Flow, Core Resonance shall execute according to the approved Challenge completion sequence.

AC-10.4.7 — Challenge Completion

The player shall be able to complete the required Challenge stages: Ring System → Alignment Analysis → Energy Flow → Core Resonance. Successful completion shall place the Challenge in its approved completion state.

AC-10.4.8 — Portal Activation

After successful Challenge completion, Portal activation shall become available and execute according to the approved completion flow.

AC-10.4.9 — Restart Behaviour

The player shall be able to restart an active Challenge. Restart shall return the Challenge to the approved Starting Chamber state and shall not treat the incomplete attempt as completed progression.

AC-10.4.10 — Challenge Exit

The player shall be able to use the approved Challenge Exit behaviour. Exiting shall not incorrectly record incomplete progress as completed.

AC-10.4.11 — Overall Acceptance Rule

Challenge Playable = PASS only when the complete approved path works without a blocking defect: Challenge Access → Starting Chamber → Ring System → Alignment Analysis → Energy Flow → Core Resonance → Challenge Completion → Portal Activation.

10.5 Rewards Granted

The reward flow shall be considered accepted when successful Challenge completion correctly produces the approved reward presentation and associated progression state.

AC-10.5.1 — Reward Eligibility

The player shall become eligible for the approved Challenge reward only after successful Challenge completion. An incomplete Challenge shall not produce reward eligibility.

AC-10.5.2 — Reward Presentation

After successful Challenge completion, the approved Reward Popup shall present the reward/result as part of the completion flow. No separate Reward Scene shall be required.

AC-10.5.3 — Reward Claimed State

The approved reward claimed state shall be correctly reflected in the player's progression data and shall not be incorrectly lost or duplicated.

AC-10.5.4 — Progression Update

Successful reward processing shall update the approved Challenge completion and reward progression state.

AC-10.5.5 — Reward Persistence

After successful reward processing, the approved reward/progression state shall be persisted through SaveManager according to the Save System.

AC-10.5.6 — Portal / Reward Flow

The approved completion flow shall execute as: Challenge Complete → Portal Activation/Warp → Reward Popup → Reward Claimed/Progression Update → HUB Return.

AC-10.5.7 — No Reward on Incomplete Challenge

An incomplete Challenge shall not be treated as completed reward progression after Restart or Exit.

AC-10.5.8 — Overall Acceptance Rule

Rewards Granted = PASS only when Successful Challenge → Reward Eligibility → Reward Presentation → Reward State → Progression Update → Persistent Save completes without a blocking defect.

10.6 Save Works

The Save System shall be considered accepted when approved player and progression data can be saved, loaded and persisted correctly without incorrectly recording incomplete Challenge progress as completed.

AC-10.6.1 — Initial Save State

A new player shall begin with Level 1, XP 0, Empty Inventory, No Completed Challenges and Forge Vision Locked.

AC-10.6.2 — Save Operation

Approved persistent player and progression data shall be saved through SaveManager.

AC-10.6.3 — Load Operation

Previously saved approved player/progression state shall be loadable and shall correctly match the saved state.

AC-10.6.4 — Challenge Completion Save

After successful Challenge completion, the approved completion and progression state shall be saved and remain available during subsequent verification.

AC-10.6.5 — Incomplete Challenge Protection

An incomplete Challenge shall not be persisted as completed progression. Exiting or restarting before successful completion shall not create completed Challenge progress.

AC-10.6.6 — Auto-Save

The approved Save System shall support auto-save through SaveManager. No specific auto-save interval shall be required here because the locked source does not define one.

AC-10.6.7 — SaveManager Ownership

Persistent saving shall remain centralized through SaveManager. SaveManager shall remain responsible for Save, Load, Auto-save and Persistent Data.

AC-10.6.8 — Overall Acceptance Rule

Save Works = PASS only when Valid State → SaveManager → Save → Load → State Verification works correctly and incomplete Challenge attempts are not persisted as completed progression.

10.7 Transitions Work

The approved scene, gameplay and UI transitions shall be considered accepted when the complete transition flow operates correctly without a blocking transition defect.

AC-10.7.1 — Opening → HUB

After the approved Opening Sequence, the player shall enter the HUB flow through the “Press ENTER to Begin” interaction and the approved HUB Awakening sequence.

AC-10.7.2 — HUB → Challenge

After the required progression, the player shall access Challenge Selection and transition from MainScene to ChallengeScene correctly.

AC-10.7.3 — Challenge Initial Transition

When ChallengeScene loads, the approved Challenge Starting Chamber state and required Challenge systems shall be available.

AC-10.7.4 — Challenge → Portal

After successful Challenge completion, Portal activation and the approved Portal transition shall execute correctly.

AC-10.7.5 — Portal → Reward Popup

After the Portal transition, the approved Reward Popup shall appear as part of the completion flow. No separate Reward Scene shall be required.

AC-10.7.6 — Reward → HUB

After the reward flow, the player shall return to MainScene/HUB through the approved Portal Warp and 1-second Black Fade.

AC-10.7.7 — Persistent Manager Continuity

SaveManager, AudioManager, PlayerManager and SettingsManager shall remain available across the approved scene transitions.

AC-10.7.8 — UI Flow Continuity

Approved UI navigation shall remain functional before and after transitions without unintended panel states, broken navigation or transition-blocking UI behaviour.

AC-10.7.9 — Overall Acceptance Rule

Transitions Work = PASS only when the approved flow executes without a blocking defect: Opening → HUB → Challenge → Portal → Reward → HUB.

10.8 Prototype Stable

The prototype shall be considered stable when the approved core flow can be executed consistently without a blocking defect that prevents required progression.

AC-10.8.1 — Core Flow Stability

The approved core prototype flow from application launch through HUB, Challenge, Reward and HUB return shall execute without a blocking defect.

AC-10.8.2 — Feature Stability

The approved major prototype features shall satisfy their required acceptance behaviour: Opening Sequence, HUB, Login/Sign-Up, Challenge, Rewards, Save System and Transitions.

AC-10.8.3 — UI Stability

Approved UI panels, popups and interactions shall remain usable throughout the required flow without broken panel states, blocking interactions or navigation failures.

AC-10.8.4 — Gameplay Stability

The approved Challenge sequence — Ring System → Alignment Analysis → Energy Flow → Core Resonance — shall remain executable through successful completion and the approved completion flow.

AC-10.8.5 — Save Stability

Approved persistent player/progression state shall reliably save and load. Successful progression shall persist and incomplete Challenge attempts shall not incorrectly persist as completed progression.

AC-10.8.6 — Transition Stability

The approved Opening → HUB → Challenge → Portal → Reward → HUB flow shall execute without a blocking transition defect.

AC-10.8.7 — Critical Defect Rule

Before final prototype acceptance, critical blocking defects shall be resolved. Prototype stability does not require a claim of zero bugs; it requires that no unresolved defect prevents approved core progression.

AC-10.8.8 — Final Stability Rule

Prototype Stable = PASS only when the approved end-to-end flow can execute consistently without a blocking defect: Launch → Opening → HUB → Login/Sign-Up → Active HUB → Challenge → Completion → Reward → Save → HUB Return.

Chapter 11 — Schedule and Budget

11.1 Development Timeline

The SkillForge Minor Project shall follow the official academic/project submission schedule provided by the college. The following milestones define the required submission and review timeline for the project.

SCH-11.1.1 — Abstract Submission & Mentor/Topic Selection

Deadline: Before 31 July 2026. The project abstract shall be submitted and the mentor/topic selection process shall be completed within the prescribed deadline.

SCH-11.1.2 — SRS Approval

Deadline: Before 15 August 2026. The Software Requirements Specification shall be prepared and submitted for approval. Approved SRS length: 72 pages.

SCH-11.1.3 — Requirement Analysis

Deadline: Before 20 September 2026. The Requirement Analysis documentation shall be completed and submitted.

Required Size: 10–15 pages.

SCH-11.1.4 — DFD, ER Diagram & Database Design

Deadline: Before 20 September 2026. Required documentation: DFD Level 0, DFD Level 1, DFD Level 2, ER Diagram and Database Design. Required Size: ≤10 pages.

SCH-11.1.5 — System Design

Deadline: Before 20 September 2026. The System Design documentation shall be completed and submitted. Required

Size: 10–15 pages.

SCH-11.1.6 — Coding & Implementation

Deadline: Before 30 November 2026. The approved SkillForge prototype shall be implemented according to the locked

SRS, architecture and project scope. College Requirement: At least 25 GitHub commits.

SCH-11.1.7 — PPT & Project Demo

Date: 05 December 2026. The completed prototype shall be presented through the required PowerPoint presentation and project demonstration.

SCH-11.1.8 — Correction & Soft Copy Approval

Date: 12 December 2026. Required corrections shall be completed and the final soft copy shall be submitted for approval.

SCH-11.1.9 — Final Viva & File Signature

Date: 15 December 2026. The final viva and required project-file signature process shall be completed.

SCH-11.1.10 — Mentor Meeting Requirement

All required mentor meetings shall be completed according to the college's prescribed project process, and the final signature requirement shall be completed on or before the applicable deadline.

SCH-11.1.11 — Official Timeline Summary

Milestone	Official Deadline
Abstract + Mentor/Topic Selection	Before 31 Jul 2026
SRS Approval	Before 15 Aug 2026
Requirement Analysis	Before 20 Sep 2026
DFD + ER Diagram + DB Design	Before 20 Sep 2026
System Design	Before 20 Sep 2026
Coding & Implementation	Before 30 Nov 2026
PPT + Project Demo	05 Dec 2026
Correction + Soft Copy Approval	12 Dec 2026
Final Viva + File Signature	15 Dec 2026

11.2 Phase-wise Progress

This section records the current completion status of the approved SkillForge development phases. Phase status shall reflect actual project progress and shall not be marked complete solely because related documentation has been prepared.

11.2.1 — Phase 1: Game Design & Core Concept

Status: 100% Complete. Completed planning includes Core Idea, Game Vision, Story Foundation, World Theme, Gameplay Loop, HUB Design, Challenge Chamber Design, Forge Vision, Skill-Ability System, Reward System, Save Logic, Audio Design, UI/UX Design, Screen Flow, State Flow, Controls, Notifications, Panel Behaviour, Hint System and Challenge Rules.

11.2.2 — Phase 2: Technical Architecture

Status: 100% Complete. Completed architecture includes Scene Architecture, Manager Architecture, Script Architecture, Prefab Architecture, Folder Structure, Data Flow Architecture and Unity Object/Scene Hierarchy.

11.2.3 — Phase 3: Software Documentation

Status: Current / In Progress. The documentation phase covers SRS, GDD, Flowcharts, DFD, ER Diagram, UML Diagrams if required and Demo Presentation Notes. The current work is the SRS documentation process.

11.2.4 — Phase 4: UI / Visual Planning

Status: Pending. Planned work includes HUB Wireframe, Panel Layouts, Analysis Panel, Login Panel, Settings Panel, Reward Popup, Notification Style, Glyph Placement, Color Palette, Fonts and Icon Pack.

11.2.5 — Phase 5: Unity Project Setup

Status: Pending. Planned setup includes Unity Version Selection, Project Creation, Folder Structure Implementation, Scene Setup, Import Settings, Git/Backup if used, Base Lighting and Camera Setup.

11.2.6 — Phase 6: Prototype Development

Status: Pending. Implementation scope includes Opening Sequence, HUB, Login System, Data Feed, Glyph System, Forge Core, Skill Tree, Portal, Challenge Chamber, Ring System, Forge Vision, Analysis Panels, Rewards, Save System, Audio, Polish and Testing.

11.2.7 — Phase 7: Finalization

Status: Pending. Final activities include Testing, Bug Fixes, Optimization, Demo Build, Presentation Rehearsal and Final Submission.

11.2.8 — Progress Status Rule

Phase status shall be updated according to actual completion of the associated work. Documentation completion shall not be treated as implementation completion for phases that require Unity development or final testing.

11.3 Tools Used

This section identifies the tools and technology categories selected for the current SkillForge prototype. Only tools supported by the locked project decisions and development requirements are treated as current project technologies.

11.3.1 — Game Engine

Unity 6.3 LTS shall be used as the primary game-development engine for the SkillForge prototype. It provides the environment required for the approved scenes, UI, gameplay, animation, audio, lighting and rendering.

11.3.2 — Programming Language

C# shall be used as the primary programming language. The approved project architecture is centered around C# Managers, Controllers, UI scripts and data classes.

11.3.3 — Unity Project Management

Unity Hub shall be used for Unity installation and project/version management.

11.3.4 — Development Environment

A compatible C# development environment shall be used for script development on the Windows development system. A specific IDE/editor product is not treated as locked in the current project documentation.

11.3.5 — Version Control

Git / GitHub shall be used for project source and version management. The project repository is established, and the college project schedule requires at least 25 GitHub commits for the Coding & Implementation milestone. Commits shall represent genuine development work.

11.3.6 — Database / Persistent Storage

SQLite shall be used as the local persistent-storage/database solution for the current prototype. The database supports the approved persistent-data requirements while SaveManager remains the sole persistent-data handler. Detailed SQL schema and implementation-level database design are outside this section and belong to the appropriate Data Management / Database Design stage.

11.3.7 — Documentation & Modelling Tools

The project documentation process shall support preparation of the SRS, GDD, DFD, ER Diagram, Flowcharts, UML diagrams if required, and presentation materials. Specific software products for diagramming or presentation are not locked and are therefore not assigned as current project tools.

11.3.8 — Operating Platform

The project shall be developed and tested on Windows 10/11 64-bit, with the current prototype targeting Windows Desktop.

11.3.9 — Future Technology Boundary

The current SRS documents only technologies required for the approved prototype. Potential future technologies such as MySQL, backend/API infrastructure, multiplayer systems, alternative programming languages, large-scale 3D expansion or AI-based systems are not current implementation technologies and are not listed as tools used by the Minor Project.

11.3.10 — Final Toolchain Summary

Category	Selected / Used
Game Engine	Unity 6.3 LTS
Programming Language	C#
Unity Management	Unity Hub
Development OS	Windows 10/11 64-bit
Version Control	Git / GitHub
Persistent Storage	SQLite
C# Development Environment	Compatible C# IDE/Editor
Documentation / Modelling	Required documentation tools; exact products not locked

11.4 Software Cost

The current SkillForge Minor Project is planned around software tools that do not require a separate paid software-license expenditure for the approved academic prototype.

11.4.1 — Unity

Unity 6.3 LTS is the selected development engine. The project follows the approved Unity Free approach for the academic prototype, resulting in no additional Unity software-license cost.

11.4.2 — C# Development

C# is the primary programming language and does not introduce a separate software-license cost for the project.

11.4.3 — Unity Hub

Unity Hub is used for Unity installation and project/version management. No separate project software cost is assigned to it under the current setup.

11.4.4 — Git / GitHub

Git/GitHub is used for source and version management. No paid subscription cost is included in the current project budget.

11.4.5 — SQLite

SQLite is selected as the local persistent-storage/database solution for the current prototype. No separate database-server licensing expenditure is required for the embedded database approach.

11.4.6 — Development System

The project assumes access to a compatible Windows development system. Existing development hardware is treated as an available project resource rather than a new software purchase.

11.4.7 — Overall Software Cost

Estimated Additional Software Cost: ₹0

This figure represents additional software licensing/subscription expenditure only. It does not claim that the complete project has zero cost when general resources such as hardware, electricity or internet are considered.

11.5 Estimated Budget

The SkillForge Minor Project is an academic prototype and is planned around existing development resources and the selected software stack. The base estimate assumes that required development hardware and general resources are already available and that currently planned software/resources do not require paid expenditure.

11.5.1 — Base Budget

Current Estimated Additional Budget: ₹0

The ₹0 figure represents the current base estimate under the existing-resource and currently planned free-resource assumption. It is not a maximum or permanent project-cost limit.

11.5.2 — Software and Development Resources

The current planned stack includes Unity, C#, Unity Hub, Git/GitHub and SQLite. No additional paid software-license expenditure is currently included in the base estimate.

11.5.3 — Existing Development Hardware

The compatible Windows development system is treated as an existing project resource. Its purchase price is therefore not included in the additional Minor Project budget.

11.5.4 — Third-Party Assets and Resources

If implementation requires paid third-party assets or resources—such as audio/SFX, VFX, video clips, transitions, models, textures, fonts, icons or other licensed materials—the actual expenditure shall be recorded separately and added to the project cost when incurred.

11.5.5 — License and Usage Verification

Before integrating any third-party asset, its applicable license and usage rights shall be verified. Where a purchase is required, the purchase/receipt record and relevant license information should be retained with the project documentation.

11.5.6 — Additional Services or Software

If a paid software tool, service, asset, hosting resource or other project-specific resource becomes necessary during implementation, its actual cost shall be recorded separately rather than being hidden within the ₹0 base estimate.

11.5.7 — Final Cost Calculation

Final Project Cost = Base Estimated Cost + Approved/Actual Additional Expenditure.

The final amount shall therefore depend on whether any paid resources are actually required and purchased during implementation.

11.5.8 — Budget Summary

Resource Category

Base Estimate

Additional Cost if Required

Current software stack

₹0

Actual cost if a paid alternative/license becomes necessary

Existing development hardware

₹0 additional

Hardware purchase only if new hardware becomes necessary

Third-party assets

₹0

Actual cost if required

Paid services / software / hosting

₹0

Actual cost if required

Total current base estimate

₹0

Not capped; depends on actual additional requirements

11.5.9 — Budget Control Rule

No paid resource shall be treated as part of the current base estimate until it is actually required for implementation. Any expenditure incurred shall be recorded with its source and applicable license/usage terms where relevant.

Chapter 12 — Appendices

12.1 Glossary

The following terms are used throughout the SkillForge SRS and describe the project's approved terminology.

Term	Meaning
SkillForge	The fantasy-and-technology-based mystery skill-development prototype.
HUB	The central Hall from which the player accesses major game functions and progression.
Challenge	A skill-based gameplay activity that the player completes for progression and rewards.
Forge Vision	The gameplay ability used to reveal and interact with Challenge analysis information.
Glyph Form	The compact default visual state of an information panel in the HUB.
Portal	The interactive world element used for Challenge entry and approved return/reward transitions.
Reward Popup	The popup UI state through which an earned Challenge reward is presented and collected.
Data Feed	The approved flow through which authenticated player data is provided to the HUB panels.
SaveManager	The Manager responsible for approved persistent player/progression data operations.
Progression	The player's accumulated advancement through levels, XP, Challenges, rewards and related approved data.
Ring System	The Challenge interaction system used for the approved formation/alignment puzzle.
Energy Flow	The approved Challenge stage that follows the required alignment.
Core Resonance	The approved Challenge stage following Energy Flow and preceding successful completion.
MainScene	The Unity scene containing the Opening, Hall, HUB and associated Main Scene functionality.
ChallengeScene	The Unity scene containing the Challenge Chamber and Challenge-specific functionality.
Manager	A centralized system component responsible for a defined area of application/game functionality.
Prefab	A reusable Unity object/template used to maintain consistency and reuse.
SQLite	The selected local database technology for persistent application data.
XP	Experience Points used as part of approved player progression.

12.2 Abbreviations

Only abbreviations relevant to the SkillForge SRS are included.

Abbreviation	Full Form / Meaning
SRS	Software Requirements Specification
GDD	Game Design Document
UI	User Interface
UX	User Experience
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
UML	Unified Modeling Language
XP	Experience Points

VFX	Visual Effects
HUB	Central gameplay/navigation area
C#	C-Sharp programming language
SQL	Structured Query Language
LTS	Long-Term Support
OS	Operating System
IDE	Integrated Development Environment
Git	Version control system used for project versioning

12.3 References

The SRS references the approved project documents and academic/project sources used as the basis for requirements, architecture, design and submission planning.

1. SkillForge SRS Master Blueprint v1.0 — Final
2. SkillForge SRS Table of Contents Specification v1.0
3. SkillForge Concept & Features / Project Decisions Reference
4. SkillForge Scene Architecture Specification v1.0
5. SkillForge Manager Architecture Specification v1.0
6. SkillForge Script Architecture Specification v1.0
7. SkillForge Prefab Architecture Specification v1.0
8. SkillForge Folder Structure Specification v1.0
9. SkillForge Data Flow / Data Management Specifications
10. SkillForge Unity Object Hierarchy & Scene Hierarchy Specification v1.0
11. SkillForge UI State Machine Specification v1.0
12. SkillForge Screen Transition Specification v1.0
13. SkillForge Panel Behaviour Specification v1.0
14. SkillForge Hint System Specification v1.0
15. Official College Minor Project Timeline / Submission Schedule
16. Minor Project Report Format provided by the college

12.4 Supporting Diagrams

Supporting diagrams are included as part of the approved project documentation set and are developed in the corresponding design and analysis documents. Detailed DFD, ERD and database design decisions are not assumed here beyond the requirements already approved in the SRS.